

Intermediary Option Pricing*

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Abstract

I exploit nights to show that intermediary inventory risk drives the option risk premium. Nights raise dealers' inventory risk on a schedule unrelated to economy-wide risk aversion, because the daily closure of regular equity markets lowers liquidity and impedes delta-hedge adjustments. I quantify dealers' "gap risk" and find that it predicts S&P 500 option returns overnight (when the friction binds) relative to the same options' day returns. After overnight equity trading emerged around 2006, loosening the friction on weeknights but not weekends, the intra-week option premium attenuated by nearly two thirds, especially in options where dealers' inventory is concentrated.

Keywords: Asset Pricing, Derivatives, Intermediation, Inventory Risk, Liquidity

JEL Codes: G10, G12, G13, G14

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Delta-hedged S&P 500 index options earn large negative average returns (Coval and Shumway, 2001; Bakshi and Kapadia, 2003). A large group of models links this option risk premium to investor preferences, including marginal utility that is convex in market returns through crash aversion, variance aversion, or disaster risk (e.g., Merton, 1976; Pan, 2002; Backus, Chernov, and Martin, 2011; Bollerslev and Todorov, 2011). A second group of models takes option demand as exogenous and focuses on option supply. In these models, dealers absorb end users' demand for downside insurance and carry the resulting short option inventory. The option premium compensates dealers for the associated inventory risk (e.g., Bollen and Whaley, 2004; Gârleanu, Pedersen, and Poteshman, 2009).

However, the link between intermediaries and asset prices is difficult to establish empirically. On the one hand, Bollen and Whaley (2004) and Gârleanu, Pedersen, and Poteshman (2009) find a positive relation between intermediary inventory and option prices, consistent with demand shocks that raise dealers' inventory and option premia. On the other hand, Chen, Joslin, and Ni (2019) find a negative relation in a more recent sample, consistent with shocks to financial constraints that reduce dealers' inventory and raise asset risk premia. The problem lies in the endogeneity of intermediary positions. Regressions of asset returns onto intermediary inventory cannot rule out that an omitted factor, for example economy-wide risk aversion, drives both intermediary positions and risk premia. Thus, the recent option pricing survey by Bates (2022) leaves unresolved the question whether intermediary inventory risk drives the option risk premium. The answer to this question is consequential beyond the options market, since option prices are routinely read as measures of investors' beliefs and risk aversion, a reading that fails if a constrained dealer is the marginal supplier (Dew-Becker and Giglio, 2026).

In this paper I show that compensation for intermediaries' inventory risk accounts for a substantial fraction of the option risk premium. To that end, I exploit two sources of variation in inventory hedging frictions: (i) the daily closure of equity markets, which raises hedging frictions over nights relative to days; and (ii) the emergence of overnight equity trading around 2006, which lowered hedging frictions over weeknights relative to weekends. Nights and weekends raise intermediary inventory risk on a schedule that is unrelated to the state of the economy. Thus, if dealers' inventory risk is compensated, then (i) dealers' inventory must predict option returns more strongly at night, when the friction binds, and (ii) after 2006 the risk premium must attenuate on weeknights relative to weekends. The

logic parallels [Du, Tepper, and Verdelhan \(2018\)](#), who trace covered interest parity deviations to quarter ends, when banks’ balance sheet constraints bind.

This paper has three findings. (i) Option dealers are exposed to elevated inventory risk over nights and weekends. I measure dealers’ daily, option-level inventory of S&P 500 options, and find that dealers’ persistent short position in out-of-the-money put options exposes them to dollar losses on their inventory, if the S&P 500 equity index drops by a large amount (about 5% or more) over nights or weekends. I refer to this inventory risk as “gap risk.” The central hedging friction is equity illiquidity. Even dealers that are delta-hedged at market close face overnight inventory risk, since overnight equity trade volume is not sufficient for them to *adjust* their delta-hedges. Dealers’ exposure to gap risk is missed by conventional (inventory gamma) measures, since gamma captures exposure to only small equity price changes. (ii) Dealer gap exposure predicts option returns more strongly overnight (when the friction binds) than during the day. (iii) After overnight equity trading emerged around 2006, loosening the friction over weeknights relative to weekends, the intra-week option risk premium attenuated by nearly two thirds relative to weekends, especially in options where dealers’ inventory position is concentrated. I conclude that a substantial fraction of the option risk premium is compensation for dealers’ inventory risk from equity illiquidity over nights and weekends.

The first part of the paper shows that dealers’ option inventory exposes them to gap risk, the risk of large overnight equity returns. I estimate dealers’ option inventory from the CBOE Open-Close data, from which I infer the intermediary sector’s daily net trades in every S&P 500 option contract. Consistent with the prior literature, I find a substantial and persistent dealer short position in options, especially in deep out-of-the-money puts. Given their short position in puts, dealers must hold a short position in stocks (or related hedge products) in order to be delta-hedged against equity market risk. Since deep out-of-the-money puts have relatively little delta, these hedging trades are relatively small on most days. However, in the event of an equity price crash, out-of-the-money puts become at-the-money puts, their delta spikes, and dealers must sell stocks in order to remain hedged. Failure to do so leads to dollar losses on their inventory.

I show that dealers’ exposure to gap risk stems from options’ “shadow gamma,” and is missed by the conventional risk measure of gamma. Option gamma measures the local curvature of the option value function and provides an estimate of exposure to small changes

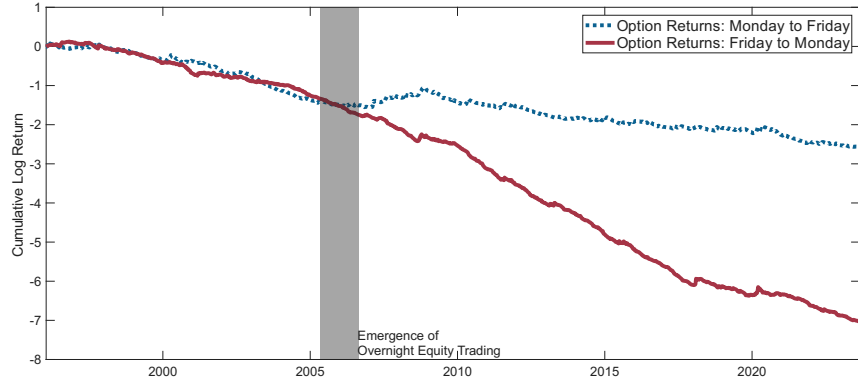
in the price of the underlying. I introduce shadow gamma as the curvature of an option, or an option portfolio, evaluated after a large hypothetical move in the underlying. Shadow gamma is a simple modification of the literature’s workhorse exposure measure and may prove useful beyond this paper. I find that option dealers’ inventory is safe from the perspective of local gamma, but substantially risky from the perspective of shadow gamma. Dealer gap exposure, the negative of the sum-product of dealers’ option positions and options’ shadow gamma, captures dealers’ exposure to gap risk.

I next compare the required hedge adjustment with observed overnight trading volume. I find that dealers’ hedging needs in the event of an equity price crash exceed typical overnight equity trading volumes. Keeping their positions delta-neutral through a 10% decline in the S&P 500 would require selling roughly \$6.3 billion of S&P 500 exposure, far larger than realized stock and futures trade volumes for most parts of the night.

The second part of the paper exploits the emergence of overnight equity trading around 2006. Several institutional changes in the mid-2000s, especially the consolidation of electronic communication networks and the expansion of electronic trading by NYSE and Nasdaq, substantially increased the ability to trade S&P 500 exposure overnight from Monday through Friday, while weekends remained largely nontradable. I exploit this market change for a difference-in-differences estimation. Close-to-close returns within the week are treated, because their overnight legs became hedgeable after 2006, while Friday-to-Monday returns (which remained unhedgeable) serve as a control. Because high-frequency option data are unavailable or unreliable over the full sample, the test uses daily close-to-close returns, which remain informative because the premium concentrates outside regular trading hours ([Jones and Shemesh, 2018](#); [Muravyev and Ni, 2020](#)). This design addresses the concern that nights differ from days for reasons other than equity illiquidity. Treated and control intervals both contain a market closure, so the comparison holds fixed whatever may be special about nontrading periods and varies whether dealers can adjust hedges during the market closure.

Figure 1 previews the results of the difference-in-differences estimation. Cumulative intra-week and weekend put returns track each other before 2006 and diverge persistently thereafter. In formal tests, intra-week put returns after 2006 become about 3.6 percentage points less negative per return interval relative to weekend returns. A triple-difference estimation shows that the effect is concentrated in short-maturity out-of-the-money puts. These are exactly the contracts in which dealers’ short position is heavily concentrated.

Figure 1: The Emergence of Overnight Equity Trading Reduced the Intra-Week Option Risk Premium Relative to Weekends



Note: The figure plots cumulative log returns of an out-of-the-money S&P 500 put portfolio, separately for weeknights and weekends. The dotted line cumulates weeknight returns (close to close on Tuesday, Wednesday, Thursday, and Friday). The solid line cumulates weekend returns (Friday close to Monday close). The puts in the portfolio have a lagged absolute delta of at most 0.5, at least two days to expiry, and are delta-hedged at the start of each return period. To make the two series comparable at a weekly frequency, each week’s weeknight observation is the simple average of the four intra-week returns. Both series are scaled to a common 10% annualized volatility over the full sample, so absolute levels are not informative. The shaded vertical band is centered on January 2006, when overnight equity trading emerged on weeknights but not weekends. Figure IA.1 in the Internet Appendix plots the cumulative difference between the two series. The sample is daily OptionMetrics data from 1996 to 2023.

This evidence shows that (i) the option risk premium attenuates when stocks become more tradable, and (ii) the attenuation is concentrated where dealer inventory is concentrated, in puts rather than calls, consistent with a reduction in dealer inventory risk.¹

The third part of the paper tests the prediction that intermediary inventory has more predictive power for option returns over periods where inventory is less hedgeable. I find that a one standard deviation increase in dealer gap exposure predicts 145 basis points lower night returns relative to the same date’s day session ($t = -3.7$). At zero measured dealer gap exposure, the estimated night versus day return differential of out-of-the-money puts is -37 basis points ($t = -0.5$), statistically indistinguishable from zero, while its sample average is 187 basis points. Thus, the book’s average measured exposure accounts for 150 of the 187 basis points.

Finally, I compare shadow gamma with the time-decay and volatility-seasonality ex-

¹Following the literature, this paper compares out-of-the-money puts to out-of-the-money calls.

planations of overnight option returns by [Jones and Shemesh \(2018\)](#) and [Muravyev and Ni \(2020\)](#). I find that these accounts and shadow gamma explain distinct components of overnight option returns. [Jones and Shemesh \(2018\)](#) show that the option risk premium is heavily concentrated over weekends, and find evidence consistent with neglect of time decay in option values. [Muravyev and Ni \(2020\)](#) show that the option risk premium is heavily concentrated over nights, and find evidence consistent with neglect of volatility seasonality. Alone, shadow gamma predicts night returns 146 basis points lower per standard deviation than the same options' day returns ($t = -7.9$). With time decay and volatility exposures held fixed, the shadow gamma slope remains -123 basis points per standard deviation ($t = -6.9$), within a sixth of its stand-alone loading.

I conclude that dealers' inventory gap risk affects the option risk premium. My findings are consistent with an interpretation where investors demand options, especially puts, to hedge against bad economic states. Dealers take the other side and end up with inventory risk from their short positions. When hedging frictions are high, the option risk premium is substantially elevated, especially in options where the dealer position is concentrated. Thus, option prices reflect intermediary inventory and hedging frictions, as well as investor beliefs and risk aversion.

Contribution. First, this paper contributes to demand-based option pricing. [Bollen and Whaley \(2004\)](#) and [Gârleanu, Pedersen, and Poteshman \(2009\)](#) show that option prices correlate positively with intermediaries' inventory, consistent with demand shocks that raise dealer inventory and risk premia. [Gruenthaler \(2023\)](#) revalues dealers' option books under crash scenarios and shows that the resulting dealer risk predicts option returns. [Chen, Joslin, and Ni \(2019\)](#) show that over a more recent sample option prices correlate negatively with public net buying-to-open trades for most months, consistent with supply shocks that lower dealer inventory capacity and raise risk premia. Thus the sign with which intermediary inventory predicts asset risk premia is not clear ex ante, and such regressions cannot directly test intermediary asset pricing models. I address this problem with a setting where the sign is clear ex ante. I show that intermediary inventory has more predictive power for option returns over periods where they face more hedging frictions, on a schedule that is unrelated to the state of the economy.

Second, this paper contributes to the broader intermediary asset pricing literature. [He and Krishnamurthy \(2013\)](#), [Adrian, Etula, and Muir \(2014\)](#), and [He, Kelly, and Manela](#)

(2017) tie risk premia across markets to intermediary capital. [Haddad and Muir \(2021\)](#) show that return predictability by intermediary distress rises with the degree of intermediation and use this pattern to derive cross-asset lower bounds, but their design does not point-identify the total effect of intermediaries or isolate the operative mechanism within a particular market. [He, Kelly, and Manela \(2017\)](#) estimate their largest price of intermediary capital risk in index options, without observing option positions. I measure option positions at the daily contract level and isolate a within-market channel based on variation in hedging frictions. Thus, I address problems of endogeneity in the study of intermediary asset pricing.

Third, the paper contributes to the literature on option returns. Delta-hedged index option returns are strongly negative, especially for puts ([Coval and Shumway, 2001](#); [Bakshi and Kapadia, 2003](#); [Broadie, Chernov, and Johannes, 2009](#); [Constantinides, Jackwerth, and Savov, 2013](#)), and concentrated over nights and weekends ([Jones and Shemesh, 2018](#); [Muravyev and Ni, 2020](#); [Bali, Goyal, Moerke, and Weigert, 2025](#)), extending to an overnight concentration of skewness and variance risk premia ([Orłowski, Schneider, and Trojani, 2024](#)), and paralleling day versus night return patterns in the underlying equity market ([Lou, Polk, and Skouras, 2019](#); [Hendershott, Livdan, and Rösch, 2020](#); [Bogouslavsky, 2021](#)). With intermediary gap risk, I provide a new economic channel for the option risk premium, one that speaks directly to the premium’s timing.

Fourth, the paper contributes to the literature on liquidity and derivative pricing, which relates option premia and crash risk to the liquidity of the underlying market ([Cao and Han, 2013](#); [Kanne, Korn, and Uhrig-Homburg, 2023](#); [Christoffersen, Feunou, Jeon, and Ornthanalai, 2021](#)). I add to this work by exploiting the emergence of overnight equity trading. This market change relaxes hedging constraints on weeknights while leaving weekends comparatively untreated, separating the effect of liquidity from aggregate changes in crash risk or volatility.

The remainder of the paper proceeds as follows. Section [I](#) measures dealers’ exposure to gap risk and introduces shadow gamma. Section [II](#) studies the effect of the emergence of overnight equity trading on option risk premia. Section [III](#) estimates the price of gap risk across options and in the state of dealers’ books, and compares shadow gamma against the competing explanations of the overnight premium. Section [IV](#) concludes. Each dataset is described in the section where it is used. Internet Appendix Section [IA.1](#) describes the S&P 500 options market, the hedging instruments, and all data sources.

I. Dealers’ Exposure to Gap Risk

This section documents dealers’ exposure to gap risk. Using CBOE trade data, I estimate dealers’ aggregate S&P 500 (SPX) option inventory and show that it embeds substantial overnight equity market risk. Dealers are persistently short out-of-the-money puts, and the equity market trading required to re hedge after a -10% underlying move far exceeds typical overnight liquidity. Dealers’ aggregate shadow gamma, the curvature of the option portfolio after a large move, is large and negative even when their local gamma is positive.

The analysis builds on work pricing dealers’ hedging frictions. [Jameson and Wilhelm \(1992\)](#) price discrete hedge rebalancing into bid-ask spreads. [Stoikov and Sağlam \(2009\)](#) show that dealers must be compensated for residual gamma and vega risk when they cannot hedge continuously. [Fournier and Jacobs \(2020\)](#) estimate a structural model of an index option dealer with limited capital. [Santa-Clara and Saretto \(2009\)](#) and [Hitzemann, Hofmann, Uhrig-Homburg, and Wagner \(2024\)](#) link the option risk premium to margin requirements and dealers’ capital constraints. The mechanism echoes classic inventory models of market making ([Ho and Stoll, 1981](#); [Grossman and Miller, 1988](#)). Dealers demand compensation for positions they cannot immediately lay off. This paper proposes gap risk, defined as inventory risk arising from limited overnight equity liquidity, as a new type of dealer risk that helps explain the timing and cross section of the option risk premium.

I.A. Dealers’ Option Inventory and Short Put Exposure

Data: Option Trade Volume by Trader Type. S&P 500 option volumes have grown to about \$2 trillion a year in premium terms in 2023 and open interest to about \$250 billion, making this the world’s largest equity options market. Because all S&P 500 index option trading occurs on a single exchange, the CBOE’s commercially available datasets permit daily measurement of the intermediary sector’s entire S&P 500 option position. This is a key data advantage of this market for the study of intermediary asset pricing. I use the CBOE Open-Close Volume files, which report daily option volumes by contract (put or call, strike, and maturity), trader group (initially firm and customer, then from 2011 onward market maker, broker-dealer, firm, customer, and professional customer), and trade direction (buy versus sell). Throughout the paper, I use the terms intermediary, dealer, and market maker interchangeably.

For the inventory measure, I measure the dealer sector’s net flow as the residual that clears the market, the negative of the combined net buys of firms and public customers. In this, I follow [Pan and Poteshman \(2006\)](#) and [Ni, Pearson, Poteshman, and White \(2021\)](#). The analysis of dealers’ exposure covers 2006 to 2023, the same window as the high-frequency pricing tests below, so dealers’ positions and the option returns they price are measured over one sample.

Beginning in 2011, the files also report market maker net flows directly, isolating the market maker category itself. These direct reports validate the residual measure. From 2011 to 2023, cumulated positions from the residual and from flagged market maker trades correlate 0.93 at the contract-day level (untabulated), and the residual series shows no discontinuity at the 2011 reporting change. Internet Appendix Table [IA.13](#) repeats the paper’s dealer pricing table on the direct reports with similar estimates.

I compute the daily net buys of the dealer sector in each option contract:

$$\text{Net Buys}_t^i = \text{Buys}_t^i - \text{Sells}_t^i, \quad (1)$$

where i indexes the option contract and t the trading day. I construct dealers’ cumulative option position by aggregating their daily net buys over the life of each contract:

$$\text{Net Position}_t^i = \begin{cases} \sum_{k=1}^t \text{Net Buys}_k^i & \text{if } t \leq \text{Expiry} \\ 0 & \text{if } t > \text{Expiry} \end{cases} \quad (2)$$

Here, k indexes trading days from the start of the sample through day t . Thus, Net Position_t^i measures the number of contracts of option i that dealers are long minus the number they are short at the end of day t . Because options are continuously listed and expire on a rolling basis, this cumulation produces a time series of dealers’ outstanding inventory after a burn-in period. The frequent expiration of derivative contracts is a major advantage when studying dealer inventory risk. In contrast to equities, where the initial position is crucial and often unobservable, options naturally reset at expiry, eliminating concerns about initial conditions or position drift. Positions cumulate from January 2005, and I treat that first year as the burn-in period, so the dealer exposure sample spans January 2006 to December 2023, matching the high-frequency return sample.

Table I: Dealer Inventory Concentrates in Short Deep Out-of-the-Money Puts

Puts		Days to Expiry		
		2 to 70	71 or more	All
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	-10.6	-2.9	-13.5
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	-2.4	-3.3	-5.6
$0.50 \leq \Delta \leq 0.75$	In-the-Money	-0.0	-0.3	-0.4
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	0.4	0.1	0.5
All		-12.6	-6.5	-19.0
Calls				
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	-2.4	-2.1	-4.5
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	1.9	1.9	3.8
$0.50 \leq \Delta \leq 0.75$	In-the-Money	1.8	0.8	2.6
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	0.7	0.1	0.8
All		2.0	0.6	2.6

Note: This table reports the average net positions of S&P 500 option dealers, sorted by moneyness and days to expiry. The net position is the number of contracts that dealers are long minus the number they are short, measured from the residual dealer net flows of the CBOE Open-Close data (see Subsection I.A). Negative values indicate that dealers are, on average, net short in the respective option category. Positions are expressed in millions of underlying equivalent units, that is, contracts multiplied by the contract multiplier of 100. Days to expiry are calendar days. Delta bins are closed at both edges, matching the construction, so an option with $|\Delta|$ exactly at a bin edge would enter both adjacent bins. Maturity buckets here are 2 to 70 and 71 or more calendar days, while the daily-data tables (Tables III and I.A.6) use 2 to 69 and 70 or more days by construction. The sample period spans January 2006 to December 2023.

Table I reports dealers' average net positions in S&P 500 options across portfolios sorted by moneyness and days to expiry. I assign options to portfolios at the daily market close and reclassify them at the next close. The dealer sector holds an aggregate short position of approximately -19.0 million underlying equivalent put units, concentrated in short-maturity deep out-of-the-money puts, where the net position reaches -10.6 million.² In call options, dealers are short the deep out-of-the-money bin (-4.5 million underlying equivalent units) and modestly long the remaining portfolios, for a small aggregate long position of $+2.6$ million. This pattern is consistent with persistent end-user demand for out-of-the-money options as crash protection, leaving dealers as the residual sellers of such options. The next subsection examines how this short put exposure translates into equity market risk for dealers.

²S&P 500 index options have a contract multiplier of 100, that is, one contract represents 100 units of the underlying index. I therefore express all option volumes and positions in underlying equivalent units by multiplying by 100.

The literature attributes dealers’ persistent short positions in S&P 500 put options largely to the hedging demands of sophisticated investors. [Lemmon and Ni \(2014\)](#) link index option trading to institutional hedging motives, noting that trading in single stock options is more often driven by retail investors. [Bollen and Whaley \(2004\)](#) report net buying pressure in index puts, consistent with institutional demand for portfolio insurance, and [Chen, Joslin, and Ni \(2019\)](#) interpret customer demand for puts as evidence of investors’ aversion to economic crash risk. Similarly, [Goyenko and Zhang \(2019\)](#) document net buying pressure in S&P 500 put options but net selling pressure in calls. Taken together, this evidence indicates that institutional investors are the primary buyers of index puts, motivated by hedging or insurance considerations, and that there is no natural counterparty to supply these contracts. Dealers therefore fill this role, resulting in a persistent short position in out-of-the-money puts.

The positions in Table I cover SPX options alone. Dealers could in principle offset part of this exposure in related markets, including options on SPY, options on E-mini futures, and over-the-counter variance contracts, but no comprehensive data on dealers’ positions across those markets exist. Such offsets would, however, only relocate the exposure. Hedging a short SPX put book requires a counterparty willing to bear the same crash risk overnight. Also, it seems unlikely that dealers absorb crash risk from customers in one index option market while selling the identical risk back to customers in a neighboring one. Measuring option demand and dealer exposure within a single options market is standard in the literature (e.g., [Gârleanu, Pedersen, and Poteshman, 2009](#); [Chen, Joslin, and Ni, 2019](#)), and [Gârleanu, Pedersen, and Poteshman \(2009\)](#) call their estimate a conservative lower bound because over-the-counter index option demand is unobserved. Thus, following this interpretation, the estimates in the subsequent sections provide a lower bound on the importance of intermediaries in the market for equity index options.

I.B. Dealers’ Liquidity Demand for Hedge Adjustments

The short put exposure documented above implies that dealers are exposed to large declines in the underlying equity index. Periods of limited equity liquidity constrain dealers’ ability to maintain delta-neutral portfolios and so expose them to equity market risk. This subsection quantifies the volume of equity trading required for dealers to remain delta-hedged when the underlying index experiences large price movements. By estimating the liquidity demand

implied by dealers' option inventory under different equity return scenarios, I show that the scale of these required hedge adjustments far exceeds actual overnight trading volumes for most hours of the night.

I estimate dealers' liquidity demand as:

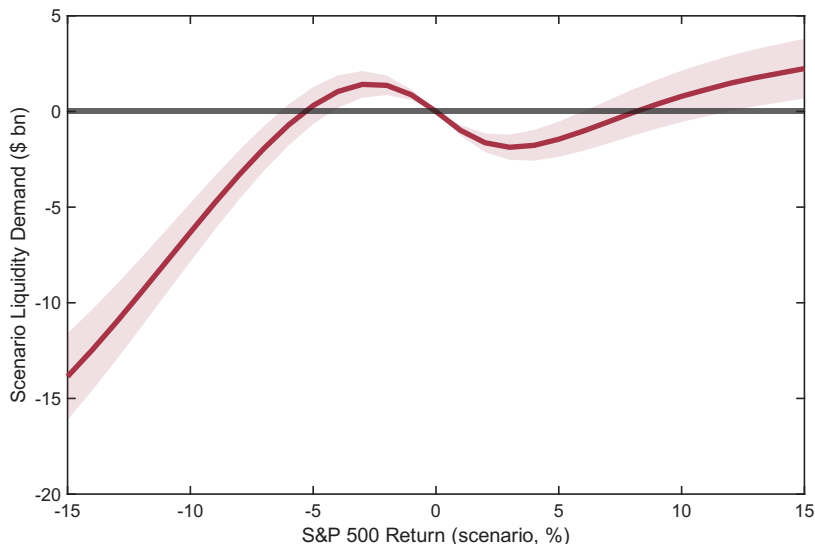
$$\text{Liquidity Demand}_{t+1}(\Delta S) = - \sum_{i=1}^I \text{Net Position}_t^i \times [\Delta_{t+1}^i(\Delta S) - \Delta_t^i] \times \text{SP500}_t, \quad (3)$$

where $\text{Liquidity Demand}_{t+1}(\Delta S)$ denotes the estimated dollar trading volume required for dealers to maintain delta-neutral positions following a hypothetical change in the underlying. The sum measures the change in the delta of dealers' book under the scenario. The hedge trade that restores delta-neutrality is its negative, so negative liquidity demand in a falling market means dealers must sell equities into the decline. Here, Net Position_t^i is dealers' net position in option i at the end of day t as defined earlier. The term Δ_t^i is the option's Black–Scholes–Merton delta at the end of day t , calculated using the option's implied volatility σ_t^i . The term $\Delta_{t+1}^i(\Delta S)$ is the estimated delta under the simulated market scenario, where $\sigma_{t+1}^i = \sigma_t^i$ and $\text{SP500}_{t+1} = \text{SP500}_t \times \mu$. The parameter μ takes values of $[0.85, 0.86, \dots, 1.15]$ to simulate S&P 500 returns of $[-15\%, -14\%, \dots, 15\%]$.

Dealers' liquidity demand for maintaining delta-hedges amounts to several billion dollars. Figure 2 plots the estimated liquidity demand from Equation (3) across hypothetical S&P 500 return scenarios. If the underlying index declines by 10%, dealers would need to sell approximately \$6.3 billion worth of equities to remain delta-hedged, and approximately \$13.9 billion after a 15% decline. This required trading volume is about ten times what the E-mini futures market transacts in a typical half hour between 20:00 and 04:00, when the underlying equities do not trade, and it is comparable to the entire realized turnover of those hours (Internet Appendix Section IA.1). Realized volume is an equilibrium quantity rather than a hard capacity limit, but it disciplines the comparison. A forced sale that absorbs many half hours of prevailing turnover could execute only at prices moved sharply against the seller. Consequently, dealers face severe constraints on hedge adjustments during these periods, exposing them to substantial inventory risk.

These estimates likely represent a lower bound. Investors' strong demand for crash protection implies that dealers hold short positions in options beyond the S&P 500 contracts examined here. Moreover, option deltas rise with expected volatility, which typically spikes

Figure 2: A -10% Gap Requires About \$6.3 Billion of Dealer Hedge Adjustments



Note: This figure plots dealers' liquidity demand for maintaining delta-hedged positions under hypothetical S&P 500 return scenarios, as defined in Equation (3). The line is the mean across days, in billions of dollars. Liquidity demand is the trading volume needed to offset the change in the book's delta. Contract deltas are converted to dollars at each day's S&P 500 closing level. Negative values on the left indicate that dealers must sell into a falling market. The shaded area shows the 95% confidence interval around the mean, computed using Newey–West standard errors with a bandwidth of 22 trading days. The sample period is January 2006 to December 2023.

during large negative equity market moves. This effect further amplifies dealers' liquidity needs relative to the estimates above, since Equation (3) keeps implied volatility constant and thus abstracts from such spikes.

I.C. Dealers' Shadow Gamma

Dealers' inventory risk is driven by options' shadow gamma, the state-contingent curvature that emerges after large underlying moves, rather than by local gamma. Gamma captures the local curvature of option values and thus how imperfect delta-hedges become for small price changes. From 2006 to 2023, however, dealers' aggregate gamma hovers around zero and, if anything, is slightly positive, implying gains from underlying price changes, not losses. This measure misses the core exposure because dealers' dominant position is short deep out-of-the-money puts. These contracts suffer large losses in a crash yet contribute little to

current gamma, since deep out-of-the-money gamma is near zero. Evaluated under large price moves, by contrast, dealers' inventories load on substantial negative shadow gamma. Positions that are well balanced for small fluctuations become acutely exposed when the underlying moves by a large amount, generating equity market risk and inventory losses, especially overnight when hedge adjustments are constrained.

Formally, consider an option with value $V(t, S)$, current underlying level S_t , and delta Δ_t . For a hypothetical underlying move ΔS , the option's shadow gamma is

$$\Gamma_t^{\text{sh}}(\Delta S) \equiv \frac{2}{(\Delta S)^2} \left[V(t, S_t + \Delta S) - V(t, S_t) - \Delta_t \Delta S \right], \quad (4)$$

so that $\frac{1}{2} \Gamma_t^{\text{sh}}(\Delta S) (\Delta S)^2$ is exactly the profit or loss of the delta-hedged position after the move. Shadow gamma coincides with local gamma as $\Delta S \rightarrow 0$, but for large moves it reflects the curvature of the option value along the entire path of the underlying. In the empirical analysis I use the tractable proxy

$$\tilde{\Gamma}_t^{\text{sh}}(\Delta S) \equiv \Gamma(t, S_t + \Delta S), \quad \text{for example } \tilde{\Gamma}_t^{\text{sh}}(-10\%) = \Gamma(t, 0.9 S_t), \quad (5)$$

which evaluates gamma at the shocked underlying level and closely tracks Γ_t^{sh} across the short-maturity out-of-the-money puts that dominate dealers' inventories.³ Internet Appendix Section [IA.3](#) derives the exact integral representation of shadow gamma, discusses how gap risk differs from jump risk, and reports average shadow gamma across option portfolios.

I aggregate options' shadow gamma to the dealer book and report the result as an *exposure*, for different scenarios of underlying S&P 500 returns:

$$\text{Dealer Gap Exposure}_{t+1}(\Delta S) = - \sum_{i=1}^I \text{Net Position}_t^i \times \tilde{\Gamma}_{i,t}^{\text{sh}}(\Delta S), \quad (6)$$

where Net Position_t^i denotes dealers' net holdings of option i at time t , and $\tilde{\Gamma}_{i,t}^{\text{sh}}(\Delta S)$ is the option's shadow gamma evaluated under a hypothetical equity market return (ΔS) . The

³Shadow gamma so measured is a one-line variation of gamma, the workhorse exposure measure of the option-pricing literature, and is therefore simple to compute and to adopt. The paper's central results go through with the exact measure of Equation (4). Internet Appendix Table [IA.15](#) reports these estimates.

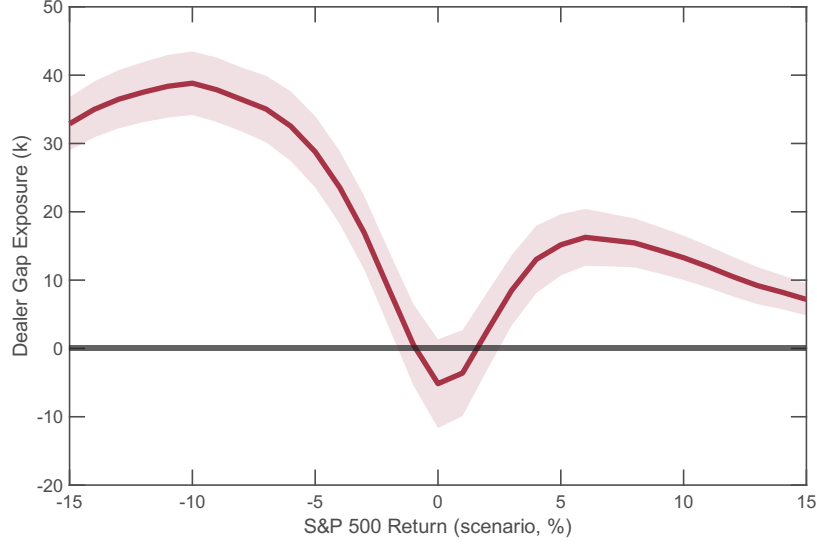
minus sign turns the book’s shadow gamma into an exposure. Dealers are net short the options whose value curves upward in a crash, so higher Dealer Gap Exposure means a book that loses more in the scenario. Dealer Gap Exposure captures the inventory risk of the whole book just as portfolio gamma, the standard curvature measure in option pricing, captures it for small moves. Gruenthaler (2023) builds an option dealer risk factor from the *slope* of the book’s value around a -10% crash. Dealer Gap Exposure is instead the *curvature* of the book’s value at the crash state, computed at constant implied volatilities. It measures the exposure that remains after delta-hedging, which is what a dealer cannot adjust overnight. I compute shadow gamma across a grid of counterfactual S&P 500 returns and report scenarios from -15% to $+15\%$. This construction links measured portfolio risk directly to the nonlinear exposure dealers face when equity liquidity constraints impede continuous reheding.

Figure 3 plots dealers’ gap exposure from Equation (6) across hypothetical S&P 500 returns. At a return of zero, the exposure is slightly negative. Dealers’ books are locally long gamma, so small equity moves generate gains rather than losses. Local measures therefore fail to capture dealers’ inventory risk over nights and weekends, when large price changes are most relevant. For large moves in either direction, however, the exposure turns positive, and it is largest around the -10% scenario, reflecting the expected losses on dealers’ short positions in deep out-of-the-money puts following a market decline.

The book’s local gamma at the prevailing index level remains positive, indicating that dealers profit from small price fluctuations while bearing risk from large ones, a pattern consistent with Hu, Kirilova, Muravyev, and Ryu (2025), who find limited delta-hedging by option dealers in Korean markets. Dealers’ portfolios are locally well balanced against small underlying moves but become increasingly exposed as price changes grow in magnitude. Inventory gamma still matters for other questions. Ni, Pearson, Poteshman, and White (2021) show that dealers’ delta-hedge rebalancing affects underlying stock prices, and Soebhag (2023) and Baltussen, Da, Lammers, and Martens (2021) conclude that option dealers’ inventory gamma captures their need to trade for delta-hedges and creates price pressure in the underlying equities.

Measuring exposure at the crash state departs from existing work. Since Gârleanu, Pedersen, and Poteshman (2009), the literature has typically estimated dealer risk using net gamma and net vega, which proxy for exposure to local price movements and to volatility

Figure 3: Dealer Inventory Is Locally Safe but Exposed to Large Declines



Note: This figure plots dealers' aggregate gap exposure at the actual and at hypothetical values of the S&P 500 index, as defined in Equation (6). The exposure is the negative of the sum across option contracts of dealers' net positions times the options' gamma evaluated at the scenario price. The line is the mean across days. Net positions are in underlying equivalent units, and the exposure is reported in thousands. Positive values indicate expected inventory losses under the corresponding equity move. The shaded area shows the 95% confidence interval around the mean, computed using Newey–West standard errors with a bandwidth of 22 trading days. The sample period is January 2006 to December 2023.

risk, respectively. Net gamma, the sum-product of dealers' net option positions and each contract's gamma, captures how much dealers' delta-hedges must adjust in response to small price movements, and it is informative when jumps are small and trading is continuous, since the underlying then rarely departs far from its current level. However, this framework cannot account for the location of the pronounced option premia. Gamma is highest for at-the-money options. If conventional gamma captured dealers' primary exposure, the most expensive contracts should be at-the-money, which the evidence rejects. Instead, observed risk premia are largest for short-maturity deep out-of-the-money puts, which is where shadow gamma is concentrated.

II. Hedging Gap Risk

This section provides the paper’s difference-in-differences evidence on the link between equity market liquidity and the option risk premium. I exploit the emergence of overnight equity trading around 2006, when the consolidation of electronic communication networks and the expansion of electronic trading by NYSE and Nasdaq made weeknight S&P 500 trading materially possible while leaving weekends largely nontradable. In a difference-in-differences setting, weeknight option returns serve as the treated group, Friday-to-Monday returns as the control. After 2006, the treatment group risk premium falls substantially relative to the control, with the largest attenuation in the out-of-the-money puts that concentrate dealers’ inventory risk, consistent with relaxed hedging constraints reducing the compensation dealers require.

II.A. Data and Delta-Hedged Option Returns

Because liquid high-frequency option quotes are substantially less available before 2006, I implement the test using daily close-to-close data, which allows for consistent measurement of option returns over the full sample from 1996 to 2023, while avoiding biases from sparse intraday observations. Thus, the difference-in-differences design uses daily S&P 500 option data from OptionMetrics, rather than the high-frequency CBOE data used in Section III. OptionMetrics aggregates option trades at the daily frequency, such that all available observations are at 16:00. The sample contains only AM-settled contracts, the standard monthly S&P 500 series. The PM-settled weekly options introduced from 2005 onward therefore never enter the panel, so the proliferation of weekly listings cannot change the sample’s composition around the 2006 event.⁴ I obtain each option’s bid quote, ask quote, and delta. While OptionMetrics applies a proprietary method for calculating options’ deltas, their deltas are typically close to Black–Scholes–Merton deltas where the volatility of the underlying is set

⁴Weekly (Friday expiry) S&P 500 options have traded since October 2005 under legacy root symbols and moved to the SPXW root in September 2011. The CBOE added Wednesday expiries in February 2016, Monday expiries in August 2016, Tuesday expiries in April 2022, and Thursday expiries in May 2022. Andersen, Fusari, and Todorov (2017) analyze the short-maturity risk information in weekly options. The growth of zero-days-to-expiry trading after 2021 (Adams, Dim, Eraker, Fontaine, Ornathanalai, and Vilkov, 2026; Beckmeyer, Branger, and Gayda, 2023) does not mechanically affect the results. The tests of this section use only AM-settled monthly contracts, and the high-frequency sample of Section III excludes contracts within two calendar days of expiry.

equal to the option’s implied volatility. To alleviate concerns of liquidity and data errors, I apply several filters to the data. I exclude options that fulfill any of the following conditions: zero trading volume on any of the previous four days, less than two days to expiry, the lagged mid quote violates the no-arbitrage lower bound, lagged mid quotes below \$0.05, lagged bid quotes below \$0.05, nonpositive lagged quoted spreads, lagged spreads above \$10, lagged spreads above the mid quote, or lagged spreads above twice the bid.

Throughout the paper, I calculate delta-hedged option returns as:

$$R_t^i = \frac{P_t^i - P_{t-1}^i - \Delta_{t-1}^i \times (\text{SP500}_t - \text{SP500}_{t-1})}{P_{t-1}^i}, \quad (7)$$

where P_t^i is the option mid quote at the end of period t , SP500 is the S&P 500 level, and Δ_{t-1}^i is the lagged option delta. This return removes first order exposure to the equity risk premium and follows the standard approach in option pricing studies (for example, [Bakshi and Kapadia, 2003](#); [Jones and Shemesh, 2018](#)). The implicit assumption that trading the underlying requires no capital is reasonable given the depth of S&P 500 futures during regular hours. Delta-hedging neutralizes small moves in the underlying but not larger ones. As prices change, deltas change, forcing continuous hedge rebalancing.

II.B. The Emergence of Overnight Equity Trading

The institutional foundations of overnight equity trading developed gradually, but the key structural changes can be traced to the mid-2000s. In the early 1990s, the New York Stock Exchange (NYSE) began limited off hours trading sessions, and by the late 1990s, regulatory and technological innovations accelerated the transition toward electronic markets. Regulation ATS (Alternative Trading Systems) of 1998 formally distinguished ATSs from registered exchanges and heightened transparency requirements. Electronic communication networks (ECNs) grew into major trading venues ([Barclay, Hendershott, and McCormick, 2003](#)), and several later consolidated and registered as full exchanges. Through this era, after-hours trading remained a small fraction of daytime activity, with substantially wider spreads ([Barclay and Hendershott, 2003, 2004](#)).

A sequence of mergers and technological upgrades followed. In 2001, the Archipelago ECN partnered with the Pacific Exchange to form ArcaEx, a fully electronic stock exchange. In April 2005, ArcaEx moved the start of its trading day, previously 08:00, to 04:00 for all U.S.

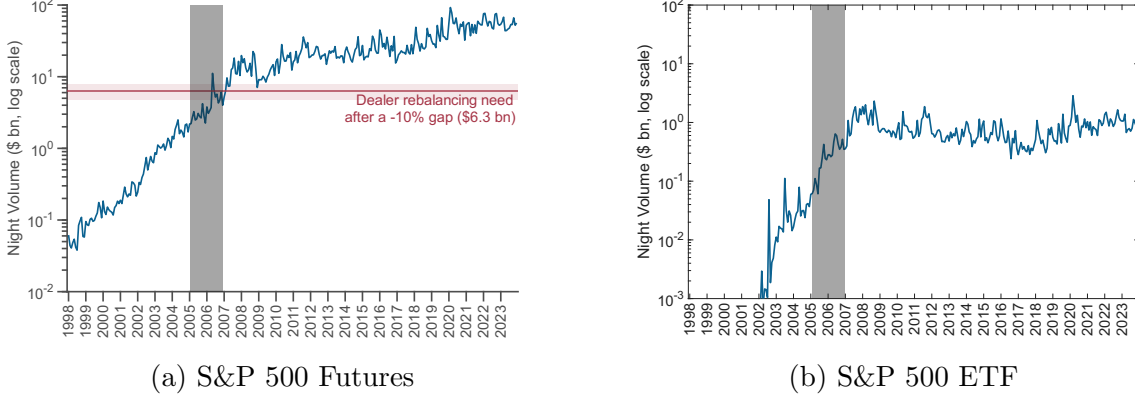
listed and OTC stocks. In the mid-2000s, the NYSE developed its Hybrid Market, combining electronic order matching with traditional floor trading (Hendershott and Moulton, 2011). The NYSE became a public company through its merger with Archipelago, completed in March 2006, and floor trading contracted sharply thereafter. Around the same time, Nasdaq, which had listed on its own market in February 2005, sought to regain competitiveness by acquiring Instinet, whose INET network traded from 07:00 to 20:00. The acquisition closed in December 2005, and Nasdaq lengthened its own session, previously 08:00 to 18:30, to run from 07:00 to 20:00 during 2006. These reforms, concentrated between April 2005 and late 2006, established the extended stock trading hours that define modern U.S. equity markets. Automation of equity trading improved liquidity in the regular session (Hendershott, Jones, and Menkveld, 2011).⁵

Consistent with these market changes, meaningful overnight equity trading activity emerged only around 2006. Although S&P 500 E-mini futures had traded nearly around the clock between Mondays and Fridays since their launch in September 1997,⁶ overnight volumes remained negligible for almost a decade. Figure 4 documents the subsequent rise in trading activity. Panel (a) plots the monthly average overnight dollar volume in the most actively traded S&P 500 E-mini futures contract against the \$6.3 billion of equity sales that a -10% move requires of dealers to remain delta-hedged (Subsection I.B). At the end of 2003, executing that single rebalancing trade would have absorbed more than six entire nights of trading. Overnight volume first covers the requirement during 2006 to 2007, and by the end of 2007 the trade fits within a fraction of a single night. Panel (b) shows the corresponding series for the SPY ETF, which grows from negligible levels in the early 2000s to roughly \$1 billion per night by 2007. While overnight ETF volumes are smaller in absolute terms, their emergence indicates that meaningful stock trading had become possible outside regular exchange hours.

⁵Sources: Regulation ATS, SEC Release No. 34-40760 (December 1998); ArcaEx’s 2001 approval as a facility of the Pacific Exchange, SEC Press Release No. 2001-122 (October 2001); ArcaEx’s 04:00 opening, SEC Release No. 34-51014 (January 10, 2005), effective April 8, 2005 (Archipelago Holdings Form 8-K, April 20, 2005); NYSE Group merger completion announcement, March 7, 2006 (SEC EDGAR, <https://www.sec.gov/Archives/edgar/data/0001326579/000089882206000141/pressmar7.txt>); Nasdaq’s February 2005 secondary offering and own market listing (SEC Form 424B4, February 2005); INET’s trading hours, Instinet Group Form 10-K for 2004; Nasdaq’s acquisition of Instinet, completed December 8, 2005 (Nasdaq Form 8-K, December 14, 2005); and Nasdaq’s session hours, SEC Release No. 34-42003 (October 1999) and SEC Release No. 34-54155 (July 14, 2006).

⁶The E-mini S&P 500 contract launched on the CME Globex electronic platform on September 9, 1997. See CME Group, “The Origins of the E-mini S&P 500,” CME Group.

Figure 4: Overnight Equity Volume First Covers Dealers' Rebalancing Need During 2006–07



Note: This figure shows overnight trading activity in S&P 500 instruments on a logarithmic scale. Panel (a) reports the monthly average dollar volume transacted overnight in the most actively traded S&P 500 E-mini futures contract. The horizontal line marks \$6.3 billion, the average equity sale dealers require to remain delta-hedged after a -10% overnight move (Subsection I.B). The six-month average of overnight volume first exceeds this requirement in April 2007. The shaded horizontal band is the 95% confidence interval for this requirement, computed using Newey–West standard errors with a bandwidth of 22 trading days. Panel (b) shows the corresponding series for the SPY ETF. Observations before 2002 fall below the \$0.001 billion axis floor. The shaded vertical band is centered on January 2006. Overnight trading is defined as transactions between 18:05 and 09:30 (E.T.). Volumes are in billions of dollars per night. The sample period is 1998 to 2023.

The reforms described above made weeknight S&P 500 trading possible while the weekend remained largely nontradable. The volume growth in Figure 4 illustrates and dates that change. Measured against hedging needs, weeknights switched from effectively nontradable to tradable within a few years centered on 2006. Dating the treatment at January 2006 is conservative. The six-month average of overnight volume enters the confidence band of the full sample hedging requirement only in mid-2006 and first exceeds the requirement itself in April 2007, so the early post period is only partially treated, which biases the estimated compression toward zero.

II.C. Risk Premia Around the Emergence of Overnight Equity Trading

The sharp increase in weeknight trading, contrasted with persistently closed weekend markets, provides a natural setting to study how equity liquidity affects the option risk premium.⁷

⁷A potential concern is that the start of overnight equity trading might have changed the overnight equity return distribution itself, and thus the option risk premium. Internet Appendix Table IA.5 and Figure IA.4

Intra-week option returns, which include nights when equity markets are tradable after 2006, serve as the treated group, while weekend returns, which span 72 hours close to close, of which about 65.5 are nontrading, serve as the control. The January 2006 break date follows the institutional record of Subsection II.B, which documents exchange reforms. The reforms stemmed from exchange competition and trading technology, so the friction loosened at a time unrelated to economy-wide risk aversion. If higher equity liquidity relaxes dealers' hedging frictions, option returns within the week should become less negative relative to those over weekends following the onset of overnight trading.

To examine how the option risk premium changes with the emergence of overnight equity trading, I estimate the following difference-in-differences specification:

$$R_t^i = \alpha + \beta_1 \text{Intra-week}_t + \beta_2 \text{Post}_t + \beta_3 \text{Intra-week}_t \times \text{Post}_t + \epsilon_t^i, \quad (8)$$

where R_t^i denotes the delta-hedged return of portfolio i of out-of-the-money S&P 500 put options.⁸ The variable Intra-week_t is an indicator for returns that fall within the week (i.e., any close-to-close return that is not Friday to Monday), and Post_t indicates observations after January 2006. I cluster standard errors by month.

Table II reports estimates of Equation (8). Before 2006, intra-week returns averaged -144.7 bps per return interval. After the emergence of overnight equity trading, the intra-week premium fell by nearly two thirds, with returns averaging -53.4 bps. The key difference-in-differences estimate, the interaction $\text{Intra-week} \times \text{Post}$, is 355.8 bps ($t = 4.05$). Intra-week returns rose relative to weekend returns once overnight equity trading emerged. The estimate is robust to winsorizing returns at the 5th and 95th percentiles and to excluding major crisis months (from August 2008 to October 2008, February 2018, and from February 2020 to April 2020). It is also essentially unchanged, at 342.2 bps ($t = 3.81$), when the lagged VIX and its interaction with Intra-week absorb volatility state differences between weekends

indicate that realized overnight S&P 500 risk shows no statistically significant change for weeknights relative to weekends around 2006. A difference-in-differences of overnight variance yields a negative interaction between Weeknight and Post, statistically indistinguishable from zero ($t = -1.2$, untabulated). The frequency of large overnight moves, if anything, rose for weeknights relative to weekends, a shift that biases against the premium compression documented below.

⁸Table II and Table III report results based on portfolios of options rather than individual contracts. Aggregating into portfolios mitigates measurement error from idiosyncratic pricing noise, thin trading, and price bounce between the bid and ask, following standard practice in the options literature (e.g., Coval and Shumway, 2001; Bakshi, Cao, and Chen, 1997).

Table II: Intra-Week Put Returns Rise Relative to Weekends After 2006

	(1) Baseline	(2) Winsorized	(3) Excl. Crashes	(4) VIX Controls	(5) 24-Hour Basis
Intra-week	86.7 (1.42)	95.0** (2.08)	86.7 (1.42)	95.6 (1.54)	-67.6** (-2.11)
Post	-264.5*** (-3.16)	-292.5*** (-5.75)	-358.0*** (-5.11)	-252.2*** (-2.91)	-88.2*** (-3.16)
Intra-week $\times$ Post	355.8*** (4.05)	304.2*** (5.33)	429.4*** (5.41)	342.2*** (3.81)	179.5*** (4.02)
Constant	-231.4*** (-4.05)	-278.8*** (-6.89)	-231.4*** (-4.05)	-239.7*** (-4.13)	-77.1*** (-4.05)
Observations	6,958	6,958	6,812	6,951	6,958
Adjusted R^2	0.01	0.02	0.01	0.01	0.00

Note: The table shows regression estimates of Equation (8), where option returns are regressed on indicators for treated periods (intra-week returns) and the post-event period (Post, observations after January 2006). Option returns are for the out-of-the-money S&P 500 put portfolio of Figure 1. Intra-week indicates all daily close-to-close returns, except Friday to Monday. Column (1) presents the baseline regression. Column (2) winsorizes returns at the 5th and 95th percentiles. Column (3) excludes the crisis months from August 2008 to October 2008, February 2018, and from February 2020 to April 2020. Column (4) adds the CBOE volatility index lagged by one day (standardized) and its interaction with the Intra-week indicator. Column (5) scales returns to a common 24-hour basis (weekend intervals span 72 hours). Columns (4) and (5) use unwinsorized returns. The VIX main effect and its interaction with Intra-week in column (4) are included but not displayed. Returns are in basis points and are delta-hedged. t -statistics are in parentheses, and standard errors are clustered by month. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is 1996 to 2023.

and weeknights (column (4)). And it keeps its statistical strength when returns are scaled to a common 24-hour basis, ruling out the mechanical horizon difference between weekend and intra-week intervals (179.5 bps per 24 hours, $t = 4.02$; column (5)). Weekend returns themselves drifted more negative over the post period, which includes the 2008 Global Financial Crisis and its broad repricing of crash risk. Of the 355.8 bps interaction, 91.3 bps come from intra-week returns improving after 2006 and 264.5 bps from weekend returns deteriorating. The implied economic magnitude is large. Had weeknights repriced like weekends, the intra-week premium from 2006 to 2023 would have averaged -409 bps rather than the observed -53 bps. The difference-in-differences estimate removes 87% of that counterfactual premium in out-of-the-money puts. The triple-difference design below absorbs aggregate and put-specific repricing common to weeknights and weekends by construction. A repricing of

crash risk specific to weekends is addressed by the crisis-month exclusions and the pre-crisis window. Ending the sample in December 2007, so that the post period predates the crisis entirely, yields a difference-in-differences of 570.1 bps ($t = 3.18$) on just two post-event years (Internet Appendix Table [IA.7](#), Panel B).

II.D. Enhancing Identification: A Triple-Difference Test

To further isolate the liquidity channel and assess where the effect is most pronounced, I extend the analysis to a triple-difference specification that interacts the treatment with cross sectional variation in option type. Specifically, I compare deep out-of-the-money puts, where dealer inventory risk is concentrated, to other option portfolios. This approach sharpens identification and reveals whether the liquidity effect operates primarily through the options that contribute most to dealers' inventory risk. The estimating equation extends Equation (8) with the deep OTM put indicator and its interactions:

$$\begin{aligned} R_t^i = & \alpha + \beta_1 \text{Intra-week}_t + \beta_2 \text{Post}_t + \beta_3 \text{Deep OTM Puts}_i \\ & + \beta_4 \text{Intra-week}_t \times \text{Post}_t + \beta_5 \text{Intra-week}_t \times \text{Deep OTM Puts}_i \\ & + \beta_6 \text{Post}_t \times \text{Deep OTM Puts}_i + \beta_7 \text{Intra-week}_t \times \text{Post}_t \times \text{Deep OTM Puts}_i + \epsilon_t^i, \end{aligned} \quad (9)$$

where Deep OTM Puts_{*i*} indicates the deep out-of-the-money put portfolio and the remaining variables are defined as in Equation (8). The coefficient of interest is β_7 . I cluster standard errors by month $\times$ portfolio.

Table [III](#) reports the estimates, computed on option returns for eight out-of-the-money portfolios (four puts and four calls). The portfolio breakpoints are defined in the table note. In column (1), the key triple interaction term, Intra-week $\times$ Post $\times$ Deep OTM Puts, is 470.9 bps ($t = 2.93$). This coefficient indicates that the improvement after 2006 in intra-week option returns documented in Table [II](#) is concentrated in deep out-of-the-money puts, the contracts that dominate dealers' short inventory positions and thus carry the highest hedging frictions. In contrast, the lower order interactions (Intra-week $\times$ Post, Post $\times$ Deep OTM Puts) are smaller in magnitude, confirming that the observed effect is not a broad repricing across all options but rather a targeted easing of the liquidity constraint faced by dealers holding downside risk. Internet Appendix Table [IA.6](#) displays lower order terms.

Columns (2) and (3) show that the triple-difference results are robust to outliers. Win-

Table III: The Return Rise After 2006 Concentrates in Deep Out-of-the-Money Puts

	(1) Baseline	(2) Winsorized	(3) Excl. Crashes	(4) VIX Controls	(5) 24-Hour Basis
Intra-week $\times$ Post $\times$ Deep OTM Puts	470.9*** (2.93)	421.3*** (4.38)	561.0*** (3.97)	466.3*** (2.81)	267.1*** (3.38)
Intra-week $\times$ Post	121.9*** (2.60)	92.9*** (2.74)	174.2*** (3.87)	106.5** (2.16)	65.7*** (2.85)
Intra-week $\times$ Deep OTM Puts	61.7 (0.56)	55.9 (0.73)	61.7 (0.56)	66.0 (0.59)	-143.2** (-2.54)
Post $\times$ Deep OTM Puts	-305.6** (-1.97)	-348.1*** (-4.13)	-409.7*** (-3.24)	-303.7* (-1.87)	-101.9** (-1.97)
Lower Order Terms	Yes	Yes	Yes	Yes	Yes
Observations	54,842	54,842	53,675	54,788	54,842
Adjusted R^2	0.00	0.01	0.01	0.00	0.00

Note: This table reports estimates of the triple-difference specification in Equation (9), in which option returns are regressed on indicators for treated options (Deep OTM Puts), treated periods (Intra-week), and the post-event period (Post, observations after January 2006). Returns are daily close to close, delta-hedged, in basis points, and computed for eight out-of-the-money S&P 500 option portfolios (four puts and four calls), formed on lagged absolute delta ($|\Delta| < 0.25$ and $0.25 \leq |\Delta| < 0.5$) and days to expiry (2 to 69 and 70 or more calendar days). Deep OTM Puts indicates the portfolio of deep out-of-the-money puts ($|\Delta| < 0.25$) with 2 to 69 days to expiry. Intra-week returns include all close-to-close observations except Friday to Monday. Column (1) presents the baseline regression. Column (2) winsorizes returns at the 5th and 95th percentiles. Column (3) excludes the crisis months from August 2008 to October 2008, February 2018, and from February 2020 to April 2020. Column (4) adds the CBOE volatility index lagged by one day (standardized) and its interactions with the Intra-week and Deep OTM Puts indicators. Column (5) scales returns to a common 24-hour basis (weekend intervals span 72 hours). Columns (4) and (5) use unwinsorized returns. The Lower Order Terms row indicates that the constant and the individual indicators are included but not displayed. The same applies to the VIX main effect and pairwise interactions in column (4). t -statistics are in parentheses, and standard errors are clustered by month $\times$ portfolio ID. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is 1996 to 2023.

sorting option returns at the 5th and 95th percentiles yields 421.3 bps ($t = 4.38$; column (2)), and excluding the crisis months from August 2008 to October 2008, February 2018, and from February 2020 to April 2020 yields 561.0 bps ($t = 3.97$; column (3)). The attenuation of the negative option risk premium is not driven by outliers or by periods of exceptional market stress. Columns (4) and (5) apply the volatility controls and the 24-hour scaling of Table II to the triple difference. The triple interaction is essentially unchanged under the VIX interactions (466.3 bps, $t = 2.81$) and strengthens statistically on the common 24-hour basis (267.1 bps, $t = 3.38$).

The inference behind the headline estimate is conservative as well. Reestimating column (1) with standard errors clustered on the time dimension alone, by date, month, or year, which allows arbitrary correlation across all eight portfolios within each time cluster, raises the t -statistic to between 3.4 and 3.8, and wild cluster bootstrap p -values for the triple interaction are 0.000 with month clusters and 0.004 with year clusters (Internet Appendix Table [IA.7](#), Panel A). The baseline month $\times$ portfolio clustering is the most conservative of these schemes.

Internet Appendix Figure [IA.2](#) traces the effect through time, plotting the difference between intra-week and weekend returns of the out-of-the-money put portfolio in two-year bins. Before 2006 the difference fluctuates around zero. An intra-week specific time trend estimated on the daily data before 2006 is 30 bps per year ($t = 1.5$, untabulated), a point estimate too imprecise to establish parallel trends on its own. Two checks address the trend concern directly. First, allowing an intra-week specific linear trend over the full sample *raises* the difference-in-differences estimate to 627.5 bps ($t = 3.95$) and the triple-difference to 768.9 bps ($t = 2.69$). The fitted differential trends are, if anything, slightly negative (-19.7 bps per year, $t = -1.9$), so no linear trend can manufacture the break (Internet Appendix Table [IA.7](#), Panel B). Second, the pattern after 2006 is a discrete level shift, already present in the 2006 to 2007 bin, not the continuation of a slope. After 2006 the difference is large and positive in every bin except 2018 to 2019 and 2020 to 2021, the two bins containing the crisis months of February 2018 and February to April 2020, where it is approximately zero. Excluding crisis months, every bin after 2006 is positive.

Nor does the January 2006 dating itself drive the estimate. Internet Appendix Figure [IA.3](#) reestimates Equation (8) with the break date moved to each January from 2002 to 2010. Every alternative dating yields a positive and statistically significant coefficient, between 213.5 and 395.0 bps around the baseline of 355.8 bps, because any break in this window splits the sample into a mostly untreated early period and a mostly treated late period.

These results do not conflict with [Muravyev and Ni \(2020\)](#), who report that unconditional overnight option returns did not attenuate as after-hours equity liquidity grew. The two tests measure different objects. A level series of night returns combines the liquidity treatment with common repricing of crash risk. The weekend control differences out that common component. Nothing in the difference-in-differences requires the level to fall, and within my sample it does not.

A potential concern for identification is that neither “weekend” nor “overnight” periods are completely nontradable. Premarket equity trading begins at 04:00 on Monday mornings, and E-mini S&P 500 futures reopen as early as Sunday 18:00. Hence, the weekend control group is only partially untreated. However, the expansion of equity trading hours around 2006 was far more pronounced for weeknights than for weekends. A typical weeknight spans 17.5 hours (from 16:00 to 09:30). Premarket trading from 04:00 alone makes about 5.5 of those hours tradable, reducing the weeknight nontradable share by roughly 30%. By contrast, a weekend period spans about 65.5 hours (17.5 plus 48), for which the same extension lowers nontradable hours by less than 10%. The core of the weekend, from the Friday close of futures trading to the Sunday evening reopening, remained closed throughout.

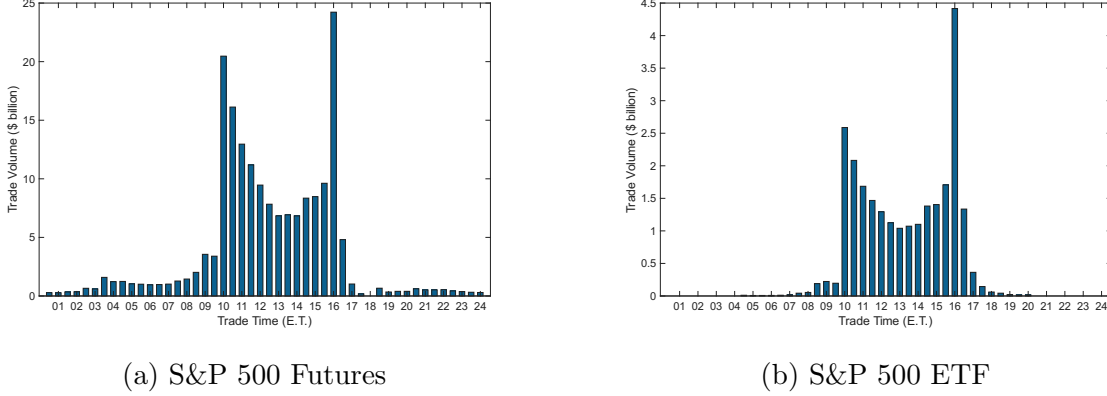
A shift in dealer positioning cannot drive these estimates either. The dealer book is approximately common to weeknights and weekends, so a general change in the level of dealers’ positions, whether around 2006 or after the Global Financial Crisis, would move both premia in the same direction.

III. The Price of Gap Risk

This section estimates the price of gap risk. I regress night and day returns of the same option contracts on the gap exposure of dealers’ books. When the book enters the night one standard deviation more exposed to a -10% gap, the returns of out-of-the-money puts fall 145 basis points further below the same date’s day returns, consistent with compensation for the risk that delta-hedges cannot be adjusted when equity markets are illiquid. The design also addresses the endogeneity of dealer inventory. As the introduction argues, a level relation between inventory and premia carries no clear sign *ex ante*, while the night versus day comparison is signed. Nights arrive independently of the state of the economy, so the friction raises dealers’ inventory risk at times unrelated to economy-wide risk aversion. Subsection [III.B](#) builds the test on that contrast.

The tests exploit the decline of equity trading volume outside of regular trading hours. Figure [5](#) plots average dollar volume in half hour intervals. S&P 500 futures trade deeply through the regular session, with roughly \$7 to \$24 billion per half hour, while volume falls by more than an order of magnitude for most of the night, and the S&P 500 ETF itself barely trades at all. Dealers can adjust delta-hedges essentially continuously during the day.

Figure 5: Equity Trading Nearly Vanishes Outside Regular Hours



Note: The figure shows the average dollar volume transacted in each 30 minute interval of the day, in billions of dollars. Times are Eastern Time. Panel (a) contains the most actively traded S&P 500 E-mini futures contract. Panel (b) contains the SPY ETF. Averages are taken across all trading days. The sample period is January 2006 to December 2023.

Overnight, the same trades face a thin market.⁹

Even if day and night equity trade volume moves with the state of the economy, the overwhelming variation in equity trade volume is between day and night. The pricing tests below therefore contrast day and night returns of the same option contracts. If compensation for gap risk reflects dealers’ constrained rebalancing, then dealers’ gap exposure should predict overnight returns relative to day returns of the same contracts. Dealers bear only the risk their hedges cannot remove, so a given book commands a larger premium when hedge adjustments are unavailable (Gârleanu, Pedersen, and Poteshman, 2009).

III.A. Data: High-Frequency Option Returns

The pricing tests in this section use high-frequency data on S&P 500 index options from the Chicago Board Options Exchange (CBOE). The dataset aggregates observations at 15-minute intervals, beginning at 09:45 E.T. (fifteen minutes after the regular market open) and

⁹S&P 500 index options themselves trade an overnight session. CBOE introduced Global Trading Hours in March 2015 and extended them to nearly around the clock in November 2021 (“Cboe to Extend Global Trading Hours for VIX and SPX Options to Nearly 24 Hours,” Cboe press release, June 15, 2021). Overnight options trading remains a small share of volume, with bid-ask spreads about twice their regular session width (Mu and Gozluklu, 2025).

ending at 16:15, the official market close. For each interval, it reports bid and ask quotes, as well as first, last, high, and low trade prices. The data further include option volume, open interest, and precomputed risk measures such as delta and gamma.¹⁰

To mitigate concerns about liquidity and data quality, I apply standard filters. To be included in the night (day) sample, an option must record positive trading volume between 16:00 and 16:15 (09:30 and 09:45), immediately prior to the return interval, with a mid quote above \$0.05 at that snapshot. I further remove entry legs with nonpositive bid or ask quotes, crossed quotes, or mid quotes below the no-arbitrage lower bound, and I exclude options within two calendar days of expiry. Finally, I exclude extreme reversals, where the mid quote changes by more than a factor of ten and reverses by more than a factor of ten across the adjacent prior close, open, and close quotes. The resulting high-frequency sample spans January 2006, when quote coverage begins, through December 2023. Tests that condition on the dealer book use the same window, matching the dealer exposure sample of Section I. With the exception of the reversal screen, which removes data errors symmetrically in both directions, all filters are based solely on lagged information and thus avoid look ahead bias. These procedures closely follow [Jones and Shemesh \(2018\)](#) and [Muravyev and Ni \(2020\)](#).

I measure night returns from 16:15 to 09:45 (E.T.) and day returns from 09:45 to 16:15. I record open prices at 09:45 because the dataset aggregates quotes into 15-minute intervals. This timing also mitigates concerns about illiquid opening quotes. Throughout the analysis, I use mid quotes as prices. I delta-hedge option returns with S&P 500 E-mini futures at the start of each period, establishing the nighttime hedge at 16:15 and holding it constant until the next morning. I estimate option deltas from the Black–Scholes–Merton model, setting the volatility of the underlying equal to the option’s lagged implied volatility. I lag implied volatility to avoid bias from the negative correlation between volatility and market returns. I report option returns in excess of the risk-free rate, which is negligible over intraday or overnight horizons and thus does not affect the results. The delta-hedged return itself is defined in Equation (7).

¹⁰A detailed description of U.S. high-frequency option data and its construction is provided in [Andersen, Archakov, Grund, Hautsch, Li, Nasekin, Nolte, Pham, Taylor, and Todorov \(2021\)](#).

III.B. The Price of Dealer Gap Exposure

Section II shows that the relative weeknight premium declined after overnight trading became feasible. I now test whether the night–day return differential varies with the measured gap exposure of dealers’ books. I compare day and night returns of the same options as the book’s exposure varies:

$$R_t^i = \beta_1 \text{Night}_t + \beta_2 \text{DGE}_{d(t)} \times \text{Night}_t + \alpha_{b(i)} + \alpha_{d(t)} + \epsilon_t^i, \quad (10)$$

where R_t^i denotes the delta-hedged return of option i , in basis points, and Night indicates overnight returns, as opposed to intraday returns. $\text{DGE}_{d(t)}$ is Dealer Gap Exposure (Section I), computed on the put book to match the put returns on the left-hand side, measured at the start of the night, and standardized to zero mean and unit variance. Internet Appendix Table IA.11 reports call-book and full-book variants. Each trading date d contributes the overnight return into its market open and the day return that follows, so the date fixed effects $\alpha_{d(t)}$ absorb everything the two periods share, including any date-level shock that jointly affects the dealer book and average return levels. One such shock is the price impact of dealer inventory that Muravyev (2016) documents at the daily frequency. Moneyiness by maturity bin fixed effects $\alpha_{b(i)}$ hold the composition of the option cross section fixed. I double cluster standard errors by trading day and option identifier. Returns are winsorized at the 2.5th and 97.5th percentiles of the pooled estimation sample, which keeps the extreme percentage returns of low-priced options from dominating the estimates.

Table IV shows regression estimates of Equation (10), on out-of-the-money S&P 500 puts, where dealers’ short gap exposure concentrates. In the baseline specification of column (1), the overnight premium moves with the state of dealers’ books. On nights on which the book enters one standard deviation more exposed, night returns are 145 bps lower relative to the same date’s day session ($t = -3.7$). Night returns run 187 bps below day returns within date on average. More dealer gap exposure means a larger dealer loss in the scenario of a large negative equity return, which is risk that dealers cannot rebalance away while equity markets are closed.

The two coefficients in column (1) measure how much of the average overnight premium the book’s gap exposure accounts for. At zero measured exposure, the fitted night versus day differential is -37 basis points ($t = -0.5$), statistically indistinguishable from zero. Zero

Table IV: Dealer Gap Exposure Predicts Lower Night Returns Within Date

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Night	-186.5*** (-4.13)	-186.7*** (-4.14)	-186.4*** (-4.14)	-187.4*** (-4.13)	-168.0*** (-3.50)	-184.9*** (-4.09)	-174.5*** (-3.83)
Dealer Gap Exposure $\times$ Night	-145.4*** (-3.74)	-142.1*** (-3.17)	-144.1*** (-3.20)	-187.0*** (-3.37)	-148.3*** (-2.72)	-146.4** (-2.39)	-171.9*** (-3.19)
Vol. controls ($\times$ Night)	No	VIX	Full	Full	Full+	Full	Full+
Trend ($\times$ Night)	No	No	No	Yes	Yes	Yes	Yes
Option chars. ($\times$ Night)	No	No	No	No	Yes	No	No
Dealer Gamma, Vanna ($\times$ Night)	No	No	No	No	No	Yes	No
Interm. constraints ($\times$ Night)	No	No	No	No	No	No	Yes
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,045,060	2,045,060	2,045,060	2,045,060	1,754,490	2,045,060	2,045,060
Adjusted R^2	0.29	0.29	0.29	0.29	0.30	0.30	0.30

Note: This table reports estimates of Equation (10). Day and night delta-hedged returns of out-of-the-money S&P 500 puts are regressed on Dealer Gap Exposure interacted with the Night indicator. Dealer Gap Exposure is the negative of the aggregate S&P 500 put book’s shadow gamma under a -10% gap (Section I), measured at the start of the night and standardized within the estimation sample. Higher values indicate a more exposed book. Returns are in basis points, winsorized at the 2.5th and 97.5th percentiles of the pooled estimation sample. All columns include date fixed effects and moneyness by maturity bin fixed effects (twenty moneyness quantiles by five maturity buckets). Each date contributes the overnight return into its market open and the day return that follows. All controls enter interacted with the Night indicator. Column (2) adds the lagged VIX. Column (3) adds λ , the lagged day versus night per-hour volatility ratio, estimated on a rolling 60-day window, and the lagged variance risk premium (“Full”). The variance risk premium is the squared VIX, converted to a daily variance, minus the trailing 22-day mean of daily realized variance from squared day and night index returns, so positive values indicate implied variance above recent realized variance. Column (4) adds a linear time trend. Column (5) adds option characteristics measured at the start of each session. The characteristics are moneyness and its square, log days to expiration and its interaction with moneyness, implied volatility, the relative quoted spread, log open interest, and log volume, plus the lagged 30 day realized volatility (“Full+”). Column (6) instead adds Dealer Gamma, the book’s local gamma, and Dealer Vanna, the sensitivity of the book’s delta to volatility, both measured on the same dealer book at the start of the night. Column (7) instead adds the intermediary capital ratio of He, Kelly, and Manela (2017) and the three month financial commercial paper to Treasury bill spread, both lagged by one trading day. t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample contains 2,045,060 day and night observations from January 2006 to December 2023. Column (5) requires nonmissing characteristics and contains 1,754,490 observations.

exposure is not an extrapolation. The put book visits zero on 340 dates in the estimation sample. The book’s sample mean lies just over one standard deviation above zero exposure. Moving the book from zero to its average therefore lowers the fitted differential by 150 basis points. At the average book, dealers’ measured gap exposure accounts for 150 of the 187 basis points of the average overnight premium of out-of-the-money puts.

Columns (2) to (7) add several control variables, and find the results to be robust. Columns (2) and (3) interact the night indicator with the lagged VIX, with λ , the day versus night volatility ratio, and with the lagged variance risk premium. The exposure slope is -144 bps ($t = -3.2$). Column (4) adds a linear time trend interacted with the night indicator, so the slope is estimated from exposure variation around trend. The estimate strengthens to -187 bps ($t = -3.4$). Column (5) adds the standard option characteristics controls interacted with the night indicator, including moneyness terms, maturity, implied volatility, the relative quoted spread, open interest, and volume, plus lagged realized volatility. The exposure slope is -148 bps ($t = -2.7$). Column (6) enters the put book’s own ordinary Greeks directly, its local gamma and its vanna. The exposure slope is -146 bps ($t = -2.4$). The night premium is associated with the book’s loss under the gap scenario rather than with its standard Greek exposures. Column (7) instead enters two intermediary constraint states directly, the [He, Kelly, and Manela \(2017\)](#) intermediary capital ratio and the financial commercial paper to Treasury bill funding spread. Neither is priced at night ($t = -0.4$ and $t = 0.9$), and the exposure slope is -172 bps ($t = -3.2$). Thus, the book’s gap exposure does not proxy for broad intermediary distress.

Price pressure around option close prices is a potential concern, but I find that this does not drive my results. A late-day demand shock could raise closing option prices and mechanically depress the following night’s return, producing a relation between the book and returns without gap risk compensation. Internet Appendix Table [IA.12](#) addresses the timing directly. The night slope is nearly unchanged when the book is measured one further trading day back (-130 bps, $t = -3.3$), so the premium reflects the persistent inventory state, not the final day’s flow. The same table shows the estimate is robust to collapsing to one night minus day return per date with Newey–West inference (-94 , $t = -4.2$) and to exact within-option matching of night and day returns (-187 , $t = -4.1$). In individual equity options, [Bali, Goyal, Moerke, and Weigert \(2025\)](#) document alternating day and night return momentum and reversal and themselves attribute the pattern to dealer capacity rather than reversing demand, showing customer order imbalances are persistent. The test here conditions on dealers’ measured book state rather than on past returns.

A further potential concern is that the book state is persistent and the sample long. A slow common drift in the book and in the night premium could masquerade as pricing. Internet Appendix Table [IA.14](#) replaces the linear trend with calendar $\times$ Night fixed effects,

which give every year, or every month, its own overnight premium, so the slope is identified only from variation of the book within the period. The estimate strengthens to -232 bps ($t = -3.9$) within years and -224 bps ($t = -2.4$) within months (-246 and -210 with the volatility controls). The book’s low frequency path works against the estimate, not for it. Internet Appendix Figure [IA.5](#), Panel (a), shows dealers’ gap exposure peaks in 2006 to 2007, is lowest in 2022, and does not grow over the sample. The decline in underlying equivalent units overstates the decline in economic exposure, because gamma per unit of the underlying falls mechanically as the index level rises. Panel (b) plots the exact dollar loss of the same book in the -10% scenario. The dollar exposure shows no downward trend, and in 2023 it returns to its 2006 level. The pricing tests use the standardized exposure, so the slope measures compensation per standard deviation of the book’s curvature.

Finally, I compare predictability across puts and calls. The Internet Appendix estimates the baseline specification segment by segment, matching each side of the dealer inventory to its own market segment (Table [IA.11](#)). Put returns are regressed on the put book’s exposure, call returns on the call book’s, and all options on the full book’s. Dealers’ call positions are net long and carry gamma that partially hedges a downside gap. Accordingly, out-of-the-money calls command no overnight premium (-62 bps, $t = -0.8$) and their own book’s exposure prices nothing (-39 bps, $t = -0.9$), while the full book estimate on all options is -109 bps ($t = -3.9$). The same table repeats all three specifications without winsorizing returns. The put book slope is -236 bps ($t = -2.7$), larger than the winsorized -145 .

III.C. Existing Explanations for the Overnight Premium

I compare shadow gamma against two prominent explanations for the overnight premium, both of which describe it as mispricing rather than compensation for risk. [Jones and Shemesh \(2018\)](#) show that option prices decay too fast over periods of market closure, as if traders priced volatility in calendar rather than trading time. Option prices then lose too much value while the market is closed and too little while it is open. Their measure is theta per dollar, the option’s rate of time decay per dollar of price. The results of [Muravyev and Ni \(2020\)](#) indicate that option traders treat a night hour like a day hour, although per-hour volatility is on average about two and a half times higher during the day. Their account carries two measures. Across options, night returns should load on convexity per dollar, $\Gamma S^2/2C$. Across dates, the premium should move with λ , the day versus night volatility

ratio.

The two option-level measures of [Jones and Shemesh \(2018\)](#) and [Muravyev and Ni \(2020\)](#) are closely related, because the variance component of theta is $\Theta^\sigma = -\sigma^2 S^2 \Gamma / 2$. One proxy, the variance component of theta per dollar, therefore carries both accounts in the horse-race. I horse-race shadow gamma against the variance component of theta per dollar (Θ^σ / C) and enter the interest component of theta per dollar separately (Θ^r / C). I add vega per dollar (V / C) so that decay competes at fixed volatility exposure, and I enter λ among the volatility controls. Here λ is the ratio of per-hour volatility during the trading day to per-hour volatility overnight, estimated for the S&P 500 on a rolling 60-day window and lagged one day, so it is known at the start of the night.

The horse-race extends the stacked design of Equation (10) to option-level proxies:

$$R_t^i = \beta_1 \text{Night}_t + \beta_2 X_t^i + \beta_3 X_t^i \times \text{Night}_t + \alpha_{b(i)} + \alpha_{d(t)} + \epsilon_t^i, \quad (11)$$

where X_t^i denotes the proxy (shadow gamma, Θ^σ / C , Θ^r / C , or V / C). Because the proxies are option-level variables, the date fixed effects $\alpha_{d(t)}$ do not absorb them, and each proxy enters through its day side level (β_2) and its interaction with the Night indicator (β_3). The joint columns of the horse-race enter all four proxies together, each with its own level and interaction pair. The sample contains both puts and calls, and the bin fixed effects $\alpha_{b(i)}$ are defined by option type, moneyness, and maturity. The volatility controls are date-level variables, so the date fixed effects absorb their levels and they enter only through their interactions with Night. The level margin is the average amount by which night returns fall below day returns, and it sits in the Night coefficient β_1 . The differential margin is how that night versus day difference varies across options, and the interaction β_3 prices it. Every proxy is standardized on the estimation sample, so slopes are basis points per standard deviation. Returns are winsorized at the 2.5th and 97.5th percentiles of the estimation sample, as in Table IV.

Table V reports the estimates. Column (1) tests the predictive power of shadow gamma, the option-level risk measure this paper introduces. Night returns fall by 146 bps per standard deviation of shadow gamma relative to the same options' day returns ($t = -7.9$). In the joint horse-race of column (4), the slope is -123 bps ($t = -6.9$), within a sixth of the solo loading, and with the volatility controls of column (5) it is -125 bps ($t = -7.1$). Ex-

Table V: Shadow Gamma Holds in the Horse-Race with Competing Explanations

	(1)	(2)	(3)	(4)	(5)
Night	-112.3*** (-3.78)	-82.8*** (-2.81)	-124.0*** (-4.19)	-109.6*** (-3.75)	-109.4*** (-3.82)
Shadow Gamma $\times$ Night	-145.8*** (-7.93)			-123.0*** (-6.93)	-124.9*** (-7.12)
$\Theta^\sigma/C \times$ Night		275.2*** (13.57)		272.4*** (11.25)	271.7*** (11.52)
$\Theta^r/C \times$ Night		-42.9 (-0.93)		-45.4 (-0.99)	-45.2 (-0.99)
$V/C \times$ Night			-164.9*** (-6.92)	-58.1** (-2.20)	-58.2** (-2.14)
Proxy levels (day side)	Yes	Yes	Yes	Yes	Yes
Vol. controls ($\times$ Night)	No	No	No	No	Full
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	3,935,656	3,935,656	3,935,656	3,935,656	3,935,656
Adjusted R^2	0.17	0.18	0.18	0.19	0.19

Note: This table reports estimates of Equation (11). Day and night delta-hedged option returns are regressed on each proxy’s level and its interaction with the Night indicator. Level coefficients are suppressed. Returns are in basis points, winsorized at the 2.5th and 97.5th percentiles of the estimation sample. Shadow Gamma is the option’s gamma at a -10% S&P 500 gap, measured at the previous close for both sessions. Θ^σ/C and Θ^r/C are the variance and interest components of theta per dollar ($\Theta^\sigma = -\sigma^2 S^2 \Gamma/2$, following Jones and Shemesh, 2018). V/C is vega per dollar. The theta and vega proxies are measured at the start of each session. All proxies are standardized on the estimation sample. All columns include date fixed effects and option type by moneyness by maturity bin fixed effects (twenty moneyness quantiles by five maturity buckets within type). Columns (1) to (3) enter Shadow Gamma, the two theta components, and vega per dollar separately. Column (4) enters all proxies jointly. Column (5) adds the lagged VIX, λ , the lagged day versus night volatility ratio, and the lagged variance risk premium, each interacted with the Night indicator (“Full”). t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample contains 3,935,656 day and night observations of S&P 500 options, 1,186,237 of them overnight returns, from January 2006 to December 2023.

cluding the crisis months of Tables II and III leaves the race slope at -123 bps ($t = -6.9$, untabulated). Within puts alone the slope is -76 bps ($t = -4.3$, untabulated), within calls -117 bps ($t = -3.3$).

Three Internet Appendix tables show the robustness of these estimates of the shadow gamma premium. Table IA.8 reestimates the stacked day versus night shadow gamma spec-

ification under alternative return measurement choices. Marking positions to executable quotes (entering at the ask, exiting at the bid), a conservative bound since actual option trades execute well inside quoted spreads (Muravyev and Pearson, 2020), strengthens the interaction of shadow gamma with Night to -217 bps ($t = -12.5$), trimming instead of winsorizing gives -166 bps, and leaving returns unhedged gives -147 bps. Restricting the sample to options with $|\Delta| \in (0.10, 0.25)$ and prices above $\$0.50$, the screen of Muravyev and Ni (2020), attenuates the estimate to -57 bps ($t = -2.3$). Table IA.9 shows that this is because both clauses of the screen remove the deep out-of-the-money options where gap risk exposure concentrates. Within the executable price sample, the average night versus day differential shrinks monotonically in magnitude from -452 bps for $|\Delta| \leq 0.05$ to -67 bps inside the delta band and -23 bps near the money, in step with each bin’s average shadow gamma, and the interaction stays significant in every bin (-138 bps, $t = -4.3$, for $|\Delta| \leq 0.05$). Table IA.10 replaces the Black–Scholes–Merton delta with a minimum-variance delta that uses the empirical smile slope to proxy the response of implied volatility to the underlying, removing the residual short delta exposure that would otherwise earn the overnight equity premium (Lou, Polk, and Skouras, 2019). Doing so lowers the average night premium by roughly an eighth and *strengthens* the interaction of shadow gamma with Night to -199 bps ($t = -6.8$).

Adding controls for time decay and volatility seasonality shows that the existing accounts and shadow gamma explain distinct components of overnight option returns. The existing explanations of the overnight option risk premium perform as described in Jones and Shemesh (2018) and Muravyev and Ni (2020). The decay proxy is the strongest option-level variable in the table. Options that lose more to variance decay per dollar earn night versus day differentials lower by 275 bps per standard deviation (column (2), $t = 13.6$), and the loading is essentially unchanged when shadow gamma enters beside it. The interest component of theta is statistically indistinguishable from zero, the split both papers predict. Vega per dollar falls from -165 bps alone (column (3)) to -58 bps in the horse-race ($t = -2.2$). The volatility ratio λ does not price the differential margin, and night returns are marginally *less* negative when λ is high, the direction opposite to the volatility split account. This date-level null speaks to the account’s time-series implication for the S&P 500. The account’s cross-sectional evidence, identified across underlyings, is not contradicted here.

The cross-sectional estimates do not by themselves distinguish mispricing from risk com-

pensation. The 2006 difference-in-differences evidence makes that distinction. The closure calendar that drives decay over nontrading periods, one-day closures on weeknights against three-day closures over weekends, was unchanged in 2006, yet the intra-week premium fell by nearly two thirds once weeknights became tradable. The attenuation holds under the volatility controls and the 24-hour scaling of Table II, columns (4) and (5), and in the pre-crisis sample (Internet Appendix Table IA.7, Panel B). The overnight premium therefore contains a component of compensation for unhedgeable gap risk that the mispricing accounts do not reach.

IV. Conclusion

Do we need intermediaries to explain the option risk premium? The link between intermediaries and asset prices is difficult to establish empirically. Intermediary asset pricing models can rationalize both positive and negative correlation between intermediary inventory and asset risk premia (Chen, Joslin, and Ni, 2019).

I exploit a setting where the prediction is clear *ex ante*. If intermediary inventory risk drives the option risk premium, then intermediary inventory should have more explanatory power for the option risk premium over periods where inventory is less hedgeable. To test this prediction I exploit comprehensive option trade data that allow for a detailed measurement of intermediary inventory, as well as two sources of variation in hedging frictions. These are the daily closure of equity markets, which raises hedging frictions over nights relative to days, and the emergence of overnight equity trading around 2006, which lowered them over weeknights relative to weekends. Nights and weekends arrive independently of the state of the economy, so these frictions move inventory risk at times unrelated to economy-wide risk aversion.

The first part of the paper shows that dealers bear inventory gap risk over nontrading periods. Dealers are on average short options, especially short-maturity deep out-of-the-money puts. Due to this inventory, dealers carry large negative shadow gamma, the curvature of the option value function evaluated at a -10% equity price move. Thus, dealers are exposed to equity price gap risk, the risk of large overnight equity price changes. This risk exposure is missed by the standard local gamma of dealer inventory.

The second part of the paper shows that the option risk premium attenuates when hedging

frictions are relaxed, especially in options where the dealer inventory is concentrated. After overnight equity trading emerged around 2006, intra-week put returns rose by 3.6 percentage points per return interval relative to weekend returns, and this change in the option risk premium is heavily concentrated in deep out-of-the-money short-maturity puts, which are the options where dealers are most short.

The third part of the paper shows that dealer gap exposure has more predictive power for option returns over nights, when inventory is less hedgeable. On nights when dealers' aggregate gap exposure enters one standard deviation higher, night returns fall 145 bps below the same date's day returns. In addition, shadow gamma, the option-level source of dealers' gap exposure, explains a component of overnight option returns that the existing mispricing-based accounts leave unexplained.

Together, the findings indicate that a substantial fraction of the option risk premium is compensation for dealers' inventory risk from equity illiquidity over nights and weekends. The fitted night premium of out-of-the-money puts is 187 basis points at the book's average gap exposure and approximately zero at zero exposure, and the intra-week premium fell by nearly two thirds once weeknights became tradable.

The results carry two implications. For market design, increasing around-the-clock equity trading can lower hedging costs and compress derivative risk premia by improving dealers' ability to manage inventory risk. For the interpretation of option prices, liquidity frictions and dealer inventories affect the price of risk, so option prices reflect these market states, as well as end investors' beliefs and risk aversion. A natural direction for future research is to develop methods that explicitly control for dealer inventory and market frictions when extracting information from option prices, for example, for tests of macro models as in [Backus, Chernov, and Martin \(2011\)](#).

References

- Adams, Greg, Chukwuma Dim, Bjorn Eraker, Jean-Sébastien Fontaine, Chayawat Ornthanalai, and Grigory Vilkov, 2026, Do S&P500 options increase market volatility? Evidence from 0DTEs, *Working Paper*.
- Adrian, Tobias, Erkki Etula, and Tyler Muir, 2014, Financial intermediaries and the cross-section of asset returns, *Journal of Finance* 69, 2557–2596.
- Andersen, Torben, Ilya Archakov, Leon Grund, Nikolaus Hautsch, Yifan Li, Sergey Nasekin, Ingmar Nolte, Manh Cuong Pham, Stephen Taylor, and Viktor Todorov, 2021, A descriptive study of high-frequency trade and quote option data, *Journal of Financial Econometrics* 19, 128–177.
- Andersen, Torben G., Nicola Fusari, and Viktor Todorov, 2017, Short-term market risks implied by weekly options, *Journal of Finance* 72, 1335–1386.
- Backus, David, Mikhail Chernov, and Ian Martin, 2011, Disasters implied by equity index options, *Journal of Finance* 66, 1969–2012.
- Bakshi, Gurdip, Charles Cao, and Zhiwu Chen, 1997, Empirical performance of alternative option pricing models, *Journal of Finance* 52, 2003–2049.
- Bakshi, Gurdip, and Nikunj Kapadia, 2003, Delta-hedged gains and the negative market volatility risk premium, *Review of Financial Studies* 16, 527–566.
- Bali, Turan G., Amit Goyal, Mathis Moerke, and Florian Weigert, 2025, In search of seasonality in intraday and overnight option returns, *Working Paper*.
- Baltussen, Guido, Zhi Da, Sten Lammers, and Martin Martens, 2021, Hedging demand and market intraday momentum, *Journal of Financial Economics* 142, 377–403.
- Barclay, Michael J., and Terrence Hendershott, 2003, Price discovery and trading after hours, *Review of Financial Studies* 16, 1041–1073.
- , 2004, Liquidity externalities and adverse selection: Evidence from trading after hours, *Journal of Finance* 59, 681–710.

- , and D. Timothy McCormick, 2003, Competition among trading venues: Information and trading on electronic communications networks, *Journal of Finance* 58, 2637–2665.
- Bates, David S., 2022, Empirical option pricing models, *Annual Review of Financial Economics* 14, 369–389.
- Beckmeyer, Heiner, Nicole Branger, and Leander Gayda, 2023, Retail traders love 0DTE options... but should they?, *Working Paper*.
- Bertsimas, Dimitris, Leonid Kogan, and Andrew W. Lo, 2000, When is time continuous?, *Journal of Financial Economics* 55, 173–204.
- Bogousslavsky, Vincent, 2021, The cross-section of intraday and overnight returns, *Journal of Financial Economics* 141, 172–194.
- Bollen, Nicolas P.B., and Robert E. Whaley, 2004, Does net buying pressure affect the shape of implied volatility functions?, *Journal of Finance* 59, 711–753.
- Bollerslev, Tim, and Viktor Todorov, 2011, Tails, fears, and risk premia, *Journal of Finance* 66, 2165–2211.
- Boyarchenko, Nina, Lars C. Larsen, and Paul Whelan, 2023, The overnight drift, *Review of Financial Studies* 36, 3502–3547.
- Boyle, Phelim P., and David Emanuel, 1980, Discretely adjusted option hedges, *Journal of Financial Economics* 8, 259–282.
- Broadie, Mark, Mikhail Chernov, and Michael Johannes, 2009, Understanding index option returns, *Review of Financial Studies* 22, 4493–4529.
- Cao, Jie, and Bing Han, 2013, Cross section of option returns and idiosyncratic stock volatility, *Journal of Financial Economics* 108, 231–249.
- Chen, Hui, Scott Joslin, and Sophie Xiaoyan Ni, 2019, Demand for crash insurance, intermediary constraints, and risk premia in financial markets, *Review of Financial Studies* 32, 228–265.

- Christoffersen, Peter, Bruno Feunou, Yoontae Jeon, and Chayawat Ornathanalai, 2021, Time-varying crash risk embedded in index options: The role of stock market liquidity, *Review of Finance* 25, 1261–1298.
- Constantinides, George M., Jens Carsten Jackwerth, and Alexi Savov, 2013, The puzzle of index option returns, *Review of Asset Pricing Studies* 3, 229–257.
- Coval, Joshua D., and Tyler Shumway, 2001, Expected option returns, *Journal of Finance* 56, 983–1009.
- Dew-Becker, Ian, and Stefano Giglio, 2026, The decline of the S&P 500 variance risk premium, *Critical Finance Review*, forthcoming.
- Du, Wenxin, Alexander Tepper, and Adrien Verdelhan, 2018, Deviations from covered interest rate parity, *Journal of Finance* 73, 915–957.
- Duffie, Darrell, Jun Pan, and Kenneth Singleton, 2000, Transform analysis and asset pricing for affine jump-diffusions, *Econometrica* 68, 1343–1376.
- Figlewski, Stephen, 1989, Options arbitrage in imperfect markets, *Journal of Finance* 44, 1289–1311.
- Fournier, Mathieu, and Kris Jacobs, 2020, A tractable framework for option pricing with dynamic market maker inventory and wealth, *Journal of Financial and Quantitative Analysis* 55, 1117–1162.
- Gârleanu, Nicolae, Lasse Heje Pedersen, and Allen M. Poteshman, 2009, Demand-based option pricing, *Review of Financial Studies* 22, 4259–4299.
- Goyenko, Ruslan, and Chengyu Zhang, 2019, Demand pressures and option returns, *Working Paper*.
- Grossman, Sanford J., and Merton H. Miller, 1988, Liquidity and market structure, *Journal of Finance* 43, 617–633.
- Gruenthaler, Thomas, 2023, Risk premia and option intermediation, *Working Paper*.

- Haddad, Valentin, and Tyler Muir, 2021, Do intermediaries matter for aggregate asset prices?, *Journal of Finance* 76, 2719–2761.
- He, Zhiguo, Bryan Kelly, and Asaf Manela, 2017, Intermediary asset pricing: New evidence from many asset classes, *Journal of Financial Economics* 126, 1–35.
- He, Zhiguo, and Arvind Krishnamurthy, 2013, Intermediary asset pricing, *American Economic Review* 103, 732–770.
- Hendershott, Terrence, Charles M. Jones, and Albert J. Menkveld, 2011, Does algorithmic trading improve liquidity?, *Journal of Finance* 66, 1–33.
- Hendershott, Terrence, Dmitry Livdan, and Dominik Rösch, 2020, Asset pricing: A tale of night and day, *Journal of Financial Economics* 138, 635–662.
- Hendershott, Terrence, and Pamela C. Moulton, 2011, Automation, speed, and stock market quality: The NYSE’s hybrid, *Journal of Financial Markets* 14, 568–604.
- Hitzemann, Steffen, Michael Hofmann, Marliese Uhrig-Homburg, and Christian Wagner, 2024, Margin requirements and equity option returns, *Working Paper*.
- Ho, Thomas, and Hans R. Stoll, 1981, Optimal dealer pricing under transactions and return uncertainty, *Journal of Financial Economics* 9, 47–73.
- Hu, Jianfeng, Antonia Kirilova, Dmitriy Muravyev, and Doojin Ryu, 2025, Options market makers, *Working Paper*.
- Jameson, Mel, and William Wilhelm, 1992, Market making in the options markets and the costs of discrete hedge rebalancing, *Journal of Finance* 47, 765–779.
- Jones, Christopher S., and Joshua Shemesh, 2018, Option mispricing around nontrading periods, *Journal of Finance* 73, 861–900.
- Kanne, Stefan, Olaf Korn, and Marliese Uhrig-Homburg, 2023, Stock illiquidity and option returns, *Journal of Financial Markets* 63, 100765.
- Leland, Hayne E., 1985, Option pricing and replication with transactions costs, *Journal of Finance* 40, 1283–1301.

- Lemmon, Michael, and Sophie Xiaoyan Ni, 2014, Differences in trading and pricing between stock and index options, *Management Science* 60, 1985–2001.
- Lou, Dong, Christopher Polk, and Spyros Skouras, 2019, A tug of war: Overnight versus intraday expected returns, *Journal of Financial Economics* 134, 192–213.
- Merton, Robert C., 1976, Option pricing when underlying stock returns are discontinuous, *Journal of Financial Economics* 3, 125–144.
- Mu, Liangyi, and Arie E. Gozluklu, 2025, Liquidity and price informativeness of options: Evidence from extended trading hours, *Journal of Futures Markets* 45, 2013–2033.
- Muravyev, Dmitriy, 2016, Order flow and expected option returns, *Journal of Finance* 71, 673–708.
- , and Xuechuan (Charles) Ni, 2020, Why do option returns change sign from day to night?, *Journal of Financial Economics* 136, 219–238.
- Muravyev, Dmitriy, and Neil D. Pearson, 2020, Options trading costs are lower than you think, *Review of Financial Studies* 33, 4973–5014.
- Ni, Sophie X., Neil D. Pearson, Allen M. Poteshman, and Joshua White, 2021, Does option trading have a pervasive impact on underlying stock prices?, *Review of Financial Studies* 34, 1952–1986.
- Orłowski, Piotr, Paul Schneider, and Fabio Trojani, 2024, On the nature of (jump) skewness risk premia, *Management Science* 70, 1154–1174.
- Pan, Jun, 2002, The jump-risk premia implicit in options: Evidence from an integrated time-series study, *Journal of Financial Economics* 63, 3–50.
- , and Allen M. Poteshman, 2006, The information in option volume for future stock prices, *Review of Financial Studies* 19, 871–908.
- Santa-Clara, Pedro, and Alessio Saretto, 2009, Option strategies: Good deals and margin calls, *Journal of Financial Markets* 12, 391–417.

Soebhag, Amar, 2023, Option gamma and stock returns, *Journal of Empirical Finance* 74, 101442.

Stoikov, Sasha, and Mehmet Sağlam, 2009, Option market making under inventory risk, *Review of Derivatives Research* 12, 55–79.

Taleb, Nassim Nicholas, 1997, *Dynamic Hedging: Managing Vanilla and Exotic Options* (John Wiley & Sons: New York).

Internet Appendix for Intermediary Option Pricing

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This Internet Appendix provides supplementary material for the paper “*Intermediary Option Pricing*.” It contains additional empirical results, robustness checks, data descriptions, and methodological details that support the main findings reported in the paper.

IA.1. Markets and Data

This section describes the S&P 500 options market, the underlying instruments used for delta-hedging, and the data sources used throughout the paper. Each dataset’s variable construction is described in the section of the paper where the dataset is used.

S&P 500 Options. This paper studies S&P 500 equity index options (SPX options), that is, put and call options written on the S&P 500 equity index of U.S. large-cap stocks. SPX options are exchange traded exclusively on the Chicago Board Options Exchange (CBOE) and were initially listed in 1983. The SPX options market is among the world’s largest and most liquid equity options markets. The high option liquidity enables a return decomposition at high frequency, and the large market size makes SPX options economically relevant to study.

The original SPX options expired once a month on that month’s third Friday. Recently, the CBOE has successively added SPXW options with different expiry dates.¹¹ I include both SPX and SPXW options in the analysis of this paper and refer to both jointly as

¹¹The expiry day timeline and the treatment of zero-days-to-expiry trading are described in Subsection [II.A](#) of the paper.

S&P 500 options. The daily analyses of Section II use AM-settled monthly contracts only (Subsection II.A). SPX options are liquid across a broad range of strike prices, which occur every \$5. Liquidity is particularly high for out-of-the-money options, which are puts (calls) with strike prices below (above) the current value of the underlying index. S&P 500 options are European and can only be exercised at expiry.

Option Dealers. Liquidity in S&P 500 options is supplied by registered market makers, the core of the dealer sector in this paper. The baseline positions of Subsection I.A measure this sector as the residual that clears the market, combining market makers with broker-dealers and professional customers. Rule 5.51 of the CBOE rulebook requires each market maker to maintain a continuous two-sided market in its appointed option classes during the trading day.¹² Dealers therefore stand ready to buy and sell throughout the trading day and absorb the resulting order flow imbalances as inventory. The CBOE does not publish the identities of its market makers. Anecdotally, they are specialized option trading firms such as Optiver, broader trading firms such as Jane Street and Citadel Securities, and some large investment banks.

S&P 500 Stocks, Futures, and ETFs. The most important delta-hedging instruments for S&P 500 options are the underlying stocks, as well as associated futures and ETFs. S&P 500 constituent stocks and SPY (the S&P 500 exchange traded fund) trade on U.S. exchanges (NYSE, Nasdaq). Regular exchange hours run from 09:30 to 16:00 (E.T.). By the mid-2000s, exchanges expanded extended hours from 04:00 to 20:00, yet there is little equity trading from 20:00 to 04:00. NYSE estimates for early 2021 put premarket trading in S&P 500 stocks at about 0.27% of total stock trading volume and postmarket trading at about 0.12%,¹³ and intraday equity volumes follow a U shape, with weekend trading closed entirely. ETFs share the same venues and hours as their underlying stocks. S&P 500 exposure via futures is provided by the CME’s E-mini contract on Globex, which trades nearly 24/5, from Sunday 18:00 to Friday 17:00, and halts daily for maintenance between

¹²Rules of Cboe Exchange, Inc., Rule 5.51 (“Market Maker Obligations”). Rule 5.52(d) specifies the standard as continuous electronic bids and offers for at least 90% of the time in at least 60% of the series of the appointed classes. https://cdn.cboe.com/resources/regulation/rule_book/C1.Exchange.Rule.Book.pdf, accessed on July 14, 2026.

¹³Stefanos Bazinas, “Earnings Announcements & Retail Participation: A deep-dive into after-hours trading,” NYSE Data Insights, May 27, 2021, <https://www.nyse.com/data-insights/earnings-announcements-and-retail-participation-a-deep-dive-into-after-hours-trading>, accessed on July 29, 2026.

17:00 and 18:00. E-mini trading supplanted the standard pit-traded contract over time. Volumes differ starkly between day and night. Average intraday E-mini dollar volume from January 2006 to December 2023 is about \$149bn per day versus about \$35bn overnight, and at higher frequency, intraday futures turnover runs from roughly \$7bn to \$24bn per 30 minute window compared with \$0.2bn to \$5bn overnight. In the hours from 20:00 to 04:00 when the underlying equities do not trade, volume averages about \$0.6bn per 30 minute window, a per-window day to night ratio of roughly 20:1 (calculated from the tick-level E-mini data described below, sampled as in [Boyarchenko, Larsen, and Whelan, \(2023\)](#)). Consistent with this, aggregate equity volumes rose from roughly \$20bn per day in the late 1990s to about \$200bn per day after 2020 (from CRSP daily volumes), but remain small in the premarket and postmarket windows. S&P 500 equity returns over day and night sessions appear in Table [IA.4](#). Equity trading volume around the clock appears in Figure 5 of the paper.

Data Sources. From CBOE, I obtain S&P 500 option prices and quotes at 15-minute intervals. Further, I obtain the daily “Open-Close Volume” files that allow for the construction of intermediary positions. From OptionMetrics, I obtain S&P 500 option prices and quotes at the daily frequency. From Refinitiv Eikon, I obtain tick-level data on S&P 500 E-mini futures traded on the CME, comprising best bid and offer quotes, trade prices, and volumes. Sampling follows standard practice, as in [Boyarchenko, Larsen, and Whelan \(2023\)](#). I obtain data on risk-free rates from the OptionMetrics IvyDB zero curve file. Data on daily stock trading volume are from CRSP. High-frequency stock volumes are from Reuters. The CBOE volatility index (VIX) is from OptionMetrics. S&P 500 index levels and special opening quotations are from Bloomberg. Commercial paper and Treasury bill rates are from the Federal Reserve Economic Data (FRED) database.

Table [IA.1](#) reports summary statistics for the paper’s main objects. The panels cover the date-level dealer book measures (Section [I](#)), delta-hedged out-of-the-money option returns by trading session, and the option-level proxies of the pricing tests (Section [III](#)).

IA.2. Day and Night Option Returns

The S&P 500 option risk premium materializes almost entirely over nontrading periods, in line with [Jones and Shemesh \(2018\)](#) and [Muravyev and Ni \(2020\)](#). In high-frequency option data from 2006 to 2023, delta-hedged overnight returns on out-of-the-money puts

Table IA.1: Summary Statistics

	Mean	SD	P5	Median	P95	N
<i>Panel A: Dealer book measures (daily, January 2006 to December 2023)</i>						
Dealer Gap Exposure	45.4	38.5	-3.5	39.4	112.6	4,529
Liquidity demand	-6.32	12.44	-28.58	-5.49	13.68	4,529
Book market value	13.9	12.8	2.6	8.4	41.4	4,529
<i>Panel B: Delta-hedged out-of-the-money option returns (basis points, 2006 to 2023)</i>						
Puts, day	44.2	3800.0	-3405.4	-111.2	3591.7	1,477,212
Puts, weeknight	-186.3	2463.5	-3301.5	-141.2	2569.7	473,077
Puts, weekend	-717.7	2729.0	-4390.6	-557.3	1860.7	125,864
Calls, day	51.6	4879.6	-5506.0	-176.2	6156.9	811,883
Calls, weeknight	-197.9	3522.2	-5009.1	-190.2	4387.4	298,536
Calls, weekend	-77.6	5097.9	-5391.4	-167.3	4730.2	73,978
<i>Panel C: Cross-sectional proxies, night pricing sample (2006 to 2023)</i>						
Shadow gamma	0.693	1.084	0.000	0.168	2.928	1,186,237
Variance decay Θ^σ/C	-47.0	69.3	-191.9	-20.2	-1.9	1,186,237
Interest decay Θ^r/C	0.021	1.369	-1.895	0.017	2.137	1,186,237
Vega per dollar V/C	17.5	14.6	3.0	13.5	47.5	1,186,237
Variance risk premium	0.390	1.573	-0.783	0.405	1.782	1,186,237

Note: The table reports summary statistics in three panels. Panel A contains the date-level dealer book measures on the residual dealer positions from January 2006 to December 2023. The measures are Dealer Gap Exposure (Section I), dealers' liquidity demand of Figure 2 evaluated at the -10% scenario, and the absolute market value of the dealer book, both in billions of U.S. dollars. Dealer Gap Exposure is computed on the put book and is in underlying equivalent units, reported in thousands, as in Panel (a) of Figure IA.5. Higher values of Dealer Gap Exposure indicate a more exposed book, and the variable is standardized where it enters regressions. For liquidity demand, negative values are required selling. Panel A weights each trading day equally. Regression estimation samples weight by option observation, so moments computed within those samples differ (Table IV). Panel B contains delta-hedged returns of out-of-the-money S&P 500 options (absolute delta of at most 0.5, more than two days to expiry) in basis points from 2006 to 2023, by option type and trading session. Weeknights are one-day closures, weekends are three-day closures, and other closure lengths enter neither of those two rows. Returns are not winsorized. The regression tables winsorize returns within their own estimation samples. Panel C contains the option-level proxies of Table V on its night estimation sample, in the units in which they enter the standardization. The proxies are shadow gamma (scaled by 1,000), the variance and interest decay proxies Θ^σ/C and Θ^r/C , vega per dollar V/C , and the lagged variance risk premium (scaled by 10,000).

and calls are both large and negative. Dealers are short the deep out-of-the-money bin of both option types, and both types earn large negative returns overnight, the pattern gap risk compensation implies. Puts, which dominate dealers' short inventory, earn the larger premium. The main text takes the day versus night facts as established by the prior literature and does not redocument them. This appendix reports the night returns in the

paper’s sample for completeness. The delta-hedged return and the high-frequency sample are defined in Subsection III.A.

Table IA.2 reports average night returns by option type, moneyness, and days to expiry. The dealer position sorts of Table I and the shadow gamma sorts of Table IA.3 use the same breakpoints. The negative overnight returns are concentrated in short-maturity, deep out-of-the-money puts and calls. Deep out-of-the-money puts with at most 70 days to expiry average -409 bps per night, and the corresponding calls average -298 bps. Consistent with dealers’ inventory, which is short the deep out-of-the-money bin of both option types but far shorter in puts, the put premium dominates. Fundamental equity risk does not generate this concentration. S&P 500 returns are less volatile overnight than during the trading day, with standard deviations of 67 basis points over weeknights and 83 over weekends, against 98 during the day (Table IA.4).

Table IA.2: Negative Night Returns Concentrate in Short-Maturity Deep Out-of-the-Money Options

		Days to Expiry		
Puts		2 to 70	71 or more	All
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	-408.9	33.5	-353.0
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	-80.2	-19.3	-65.1
$0.50 \leq \Delta \leq 0.75$	In-the-Money	-25.5	-17.8	-21.5
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	-34.7	-32.0	-34.0
All		-289.9	3.5	-246.8
Calls				
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	-297.7	8.7	-269.5
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	-112.9	-31.0	-103.3
$0.50 \leq \Delta \leq 0.75$	In-the-Money	-71.9	-26.9	-62.1
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	-49.5	-7.1	-46.8
All		-204.1	-12.3	-180.3

Note: The table shows average S&P 500 option returns for sixteen portfolios (eight puts and eight calls), sorted by moneyness and days to expiry. The All column aggregates across maturities. Returns are measured from option market close at 16:15 to the subsequent market open at 09:45. Returns are in basis points and are delta-hedged at the beginning of the respective period. Days to expiry are calendar days. Delta bins are closed at both edges, matching the construction, so an option with $|\Delta|$ exactly at a bin edge would enter both adjacent bins. The sample period is 2006 to 2023.

IA.3. Gap Risk and Options' Shadow Gamma

This section discusses the riskiness of options that stems from equity price gap risk. Gap risk is the risk of price changes over periods where liquidity is low or absent. Equity price changes over nights or weekends are natural examples. Equity price gap risk contributes to option risk, since delta-hedges can be adjusted only imperfectly over periods where the underlying is illiquid. For large gaps, local gamma understates the relevant curvature, while shadow gamma incorporates the curvature encountered after the underlying has moved.

IA.3.A. Gap Risk

Gap risk, the risk of price changes over periods where liquidity is low, is frequently discussed by traders. First, and most importantly, gaps differ from jumps in their magnitude. Intraday jumps identified in high-frequency data are typically small, while equity returns over nights or weekends can amount to several percent. Table [IA.4](#) shows S&P 500 returns over days, nights, weekends, and longer market closures. For this paper, the left return tail over nights and weekends matters most. The worst night return since 1996 is -7.5% , and the worst weekend return -12.4% (Table [IA.4](#)). The closure split in the table also shows that realized gap risk does not scale with the calendar length of the closure. Return dispersion rises only mildly from one-day nights to closures of four or more days, and the worst gap arrived over a regular weekend, not over the seven day September 2001 closure. Returns of these magnitudes are not well conceptualized as small jumps and, as I argue below, options' exposure to such gaps is not well approximated via gamma.

A second difference between gaps and jumps is that the timing of the former is typically predictable. Except around prescheduled economic announcements, jumps tend to occur at random times and tend to be modeled with some probability of arrival ([Merton, 1976](#); [Duffie, Pan, and Singleton, 2000](#)). The window of a gap is scheduled in advance, since every market close opens one, while the size of the move is not. Risk therefore lies in the magnitude of the gap rather than its timing.

IA.3.B. Gamma and Shadow Gamma

Gamma, Small Jumps, and Delta-Hedged P&L. Options' exposure to jumps is typically approximated via gamma (Γ). Gamma is the second order derivative of the option price with respect to the price of the underlying. As such, Γ captures the *local* curvature or convexity of the option value function. While the slope of the option value function can be hedged against equity price changes with a static equity position, the curvature cannot. The hedging errors that discrete rebalancing leaves behind are classical (Boyle and Emanuel, 1980; Leland, 1985; Figlewski, 1989; Bertsimas, Kogan, and Lo, 2000).

Consider an option with value $V(t, S)$, delta $\Delta_t = \partial_S V(t, S_t)$, and gamma $\Gamma_t = \partial_{SS}^2 V(t, S_t)$. A dealer who delta-hedges holds the option and $-\Delta_t$ units of the underlying, so the value of the hedged portfolio is

$$\Pi_t = V(t, S_t) - \Delta_t S_t. \quad (\text{IA.1})$$

Over a short interval $[t, t+h]$ the underlying may experience a price change $\Delta S = S_{t+h} - S_t$. The change in the value of the hedged portfolio due to this move is

$$\Delta \Pi(\Delta S) = V(t+h, S_t + \Delta S) - V(t, S_t) - \Delta_t \Delta S. \quad (\text{IA.2})$$

For a small move ΔS , use a second order Taylor expansion around (t, S_t) :

$$V(t+h, S_t + \Delta S) \approx V(t, S_t) + \partial_t V(t, S_t) h + \Delta_t \Delta S + \frac{1}{2} \Gamma_t (\Delta S)^2. \quad (\text{IA.3})$$

Plugging this into (IA.2) and canceling the first order term in ΔS yields

$$\Delta \Pi \approx \partial_t V(t, S_t) h + \frac{1}{2} \Gamma_t (\Delta S)^2. \quad (\text{IA.4})$$

Ignoring the smooth time decay term $\partial_t V(t, S_t) h$, the jump induced P&L of the delta-hedged position is

$$\Delta \Pi(\Delta S) \approx \frac{1}{2} \Gamma_t (\Delta S)^2. \quad (\text{IA.5})$$

Equation (IA.5) shows why gamma is the standard measure of option exposure to small

jumps. Once delta risk has been hedged out, the remaining exposure to small price moves is proportional to Γ_t and scales with the square of the jump size. An options portfolio with a positive gamma gains in value from ΔS when hedges remain unadjusted.

Shadow Gamma and Large Gaps. For larger gaps ΔS the local approximation in (IA.3) becomes inaccurate, because the curvature (Γ) of V changes along the move. The delta-hedged P&L for any finite jump can be written as

$$\Delta\Pi(\Delta S) = V(t, S_t + \Delta S) - V(t, S_t) - \Delta_t \Delta S. \quad (\text{IA.6})$$

This can be expressed in gamma form

$$\Delta\Pi(\Delta S) = \frac{1}{2} \Gamma_t^{\text{sh}}(\Delta S) (\Delta S)^2, \quad (\text{IA.7})$$

by defining the *shadow gamma* for price move size ΔS as

$$\Gamma_t^{\text{sh}}(\Delta S) \equiv \frac{2}{(\Delta S)^2} \left[V(t, S_t + \Delta S) - V(t, S_t) - \Delta_t \Delta S \right]. \quad (\text{IA.8})$$

By construction, (IA.7) is an identity for any ΔS .

By integrating the option's gamma along the price path from S_t to $S_t + \Delta S$, $\Gamma_t^{\text{sh}}(\Delta S)$ can be written exactly as

$$\Gamma_t^{\text{sh}}(\Delta S) = 2 \int_0^1 (1 - u) \Gamma(t, S_t + u \Delta S) du. \quad (\text{IA.9})$$

That is, shadow gamma is a weighted average of the option's future curvature along the path from S_t to $S_t + \Delta S$. Shadow gamma captures the exact change in the value of a delta-hedged options portfolio for large changes in the price of the underlying.

Note that $\lim_{\Delta S \rightarrow 0} \Gamma_t^{\text{sh}}(\Delta S) = \Gamma_t$, so that shadow gamma coincides with local gamma for small changes in S , but for large S changes $\Gamma_t^{\text{sh}}(\Delta S)$ loads on the curvature of $V(\cdot)$ over the entire change of S .

For deep out-of-the-money puts, Γ_t can be close to zero, even though a large downward move would take the option close to the money, where curvature is large. Local gamma therefore understates the overnight risk of such positions, whereas $\Gamma_t^{\text{sh}}(\Delta S)$ remains large in

magnitude because it incorporates the future curvature after a sizeable move.

Equation (IA.9) defines the exact shadow gamma $\Gamma_t^{\text{sh}}(\Delta S)$ as a weighted average of local gammas along the entire price path from S_t to $S_t + \Delta S$. In the empirical analysis I use a tractable proxy,

$$\tilde{\Gamma}_t^{\text{sh}}(\Delta S) \equiv \Gamma(t, S_t + \Delta S), \quad (\text{IA.10})$$

and in particular

$$\tilde{\Gamma}_t^{\text{sh}}(-10\%) = \Gamma(t, 0.9S_t). \quad (\text{IA.11})$$

For the short-maturity deep out-of-the-money puts that dominate dealers' inventories, $\Gamma(t, S)$ is negligible at S_t and along most of the path, and builds up only once the index has fallen close to the strike, so the curvature at the shocked underlying governs how the integral in (IA.9) varies across these contracts. Evaluating gamma at the shocked underlying therefore provides a simple and economically meaningful proxy for the theoretical shadow gamma. In what follows, I refer to $\tilde{\Gamma}_t^{\text{sh}}$ simply as shadow gamma.¹⁴

$\tilde{\Gamma}_t^{\text{sh}}(-10\%)$ is the curvature of the option value function (both hedged and unhedged) in the scenario of a -10% underlying return. An option portfolio's shadow gamma is the sum-product of the option positions (in number of options) and the respective options' individual shadow gamma. Thus, an option with a shadow gamma of 1.4×10^{-3} , the average for short-maturity deep out-of-the-money puts in Table IA.3, adds 1.4×10^{-3} units of curvature to a portfolio. The pricing regressions in the paper standardize option-level shadow gamma, so the estimated slopes are basis points of return per standard deviation of this curvature in the option cross section, rather than per unit of curvature.

Table IA.3 shows the average $\tilde{\Gamma}_t^{\text{sh}}(-10\%)$ across option portfolios over the sample. The properties of gamma naturally translate to the properties of $\tilde{\Gamma}_t^{\text{sh}}(-10\%)$, such as strict positivity for options. Shadow gamma is heavily concentrated in short-maturity deep out-of-the-money puts, with a value of 1.4×10^{-3} , versus a value of 0.2×10^{-3} for short-maturity in-the-money puts. A value of 1.4×10^{-3} indicates that buying one short-maturity deep out-

¹⁴Taleb (1997) uses the same term for a different object, a local gamma computed with the response of implied volatility to the spot move. Shadow gamma here holds volatility fixed and evaluates the curvature at the crash state.

Table IA.3: Shadow Gamma Concentrates in Short-Maturity Deep Out-of-the-Money Puts

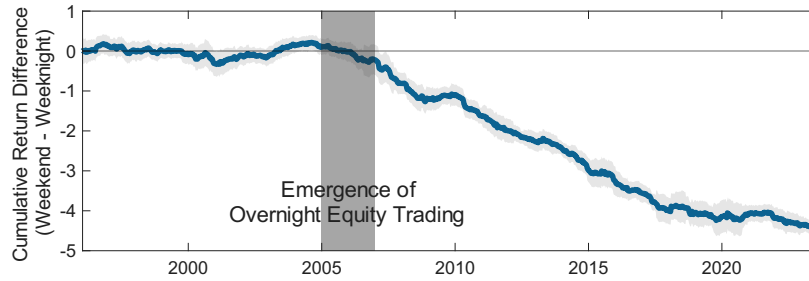
Puts		Days to Expiry		
		2 to 70	71 or more	All
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	1.4	0.6	1.2
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	0.6	0.9	0.8
$0.50 \leq \Delta \leq 0.75$	In-the-Money	0.2	0.6	0.3
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	0.0	0.2	0.1
All		1.1	0.7	1.0
Calls				
$0.00 \leq \Delta \leq 0.25$	Deep Out-of-the-Money	0.0	0.1	0.0
$0.25 \leq \Delta \leq 0.50$	Out-of-the-Money	0.2	0.6	0.3
$0.50 \leq \Delta \leq 0.75$	In-the-Money	0.5	0.9	0.6
$0.75 \leq \Delta \leq 1.00$	Deep In-the-Money	1.2	0.9	1.1
All		0.3	0.5	0.4

Note: The table shows average option-level shadow gamma for the sixteen portfolios of Table IA.2, sorted by moneyness and days to expiry. The All column aggregates across maturities. Shadow gamma is the option's gamma evaluated at a hypothetical -10% underlying return and is multiplied by 1000 for readability. Days to expiry are calendar days. Delta bins are closed at both edges, matching the construction, so an option with $|\Delta|$ exactly at a bin edge would enter both adjacent bins. The sample period is 2006 to 2023.

of-the-money put on average adds a curvature of 1.4×10^{-3} units at -10% to an investor's portfolio value function.

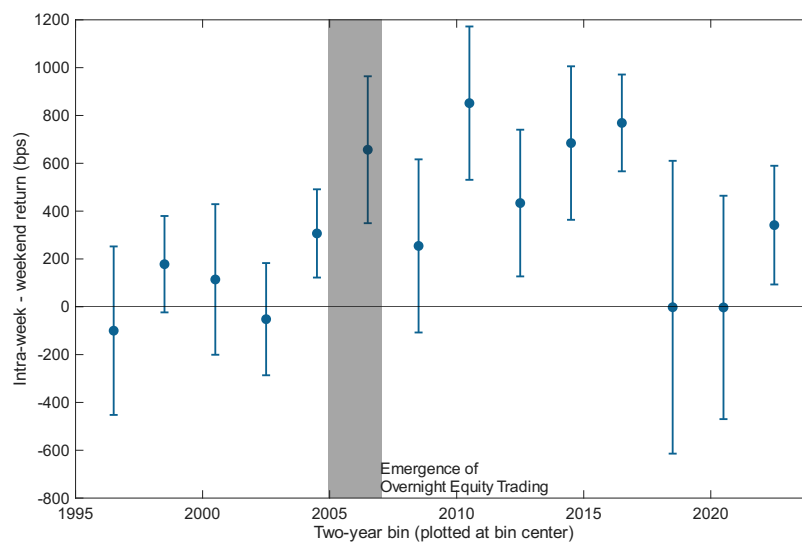
IA.4. Additional Figures

Figure IA.1: The Cumulative Weekend Minus Weeknight Return Difference Falls Persistently After 2006



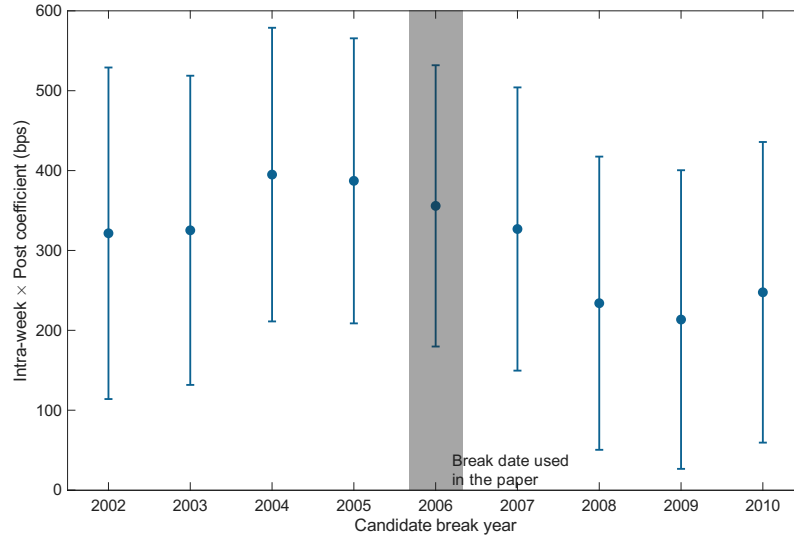
Note: The figure plots the cumulative difference between weekend and weeknight log returns of the out-of-the-money S&P 500 put portfolio of Figure 1 (puts with a lagged absolute delta of at most 0.5, at least two days to expiry, delta-hedged at the start of each return period). Negative values mean that weeknight put returns are less negative than weekend put returns. Weekend returns are Friday close to Monday close. Weeknight returns are the remaining close-to-close returns, averaged within each week so that both series are weekly. Each series is scaled to a common 10% annualized volatility over the full sample before differencing. The shaded area shows the 95% confidence interval for one year of accumulation, computed from a centered rolling standard deviation over 52 weeks of the weekly difference. The shaded vertical band is centered on January 2006, when overnight equity trading emerged on weeknights but not weekends. Figure 1 in the main text plots the two series separately. The sample is daily OptionMetrics data from 1996 to 2023.

Figure IA.2: The Intra-Week Versus Weekend Return Difference Widens Sharply After 2006



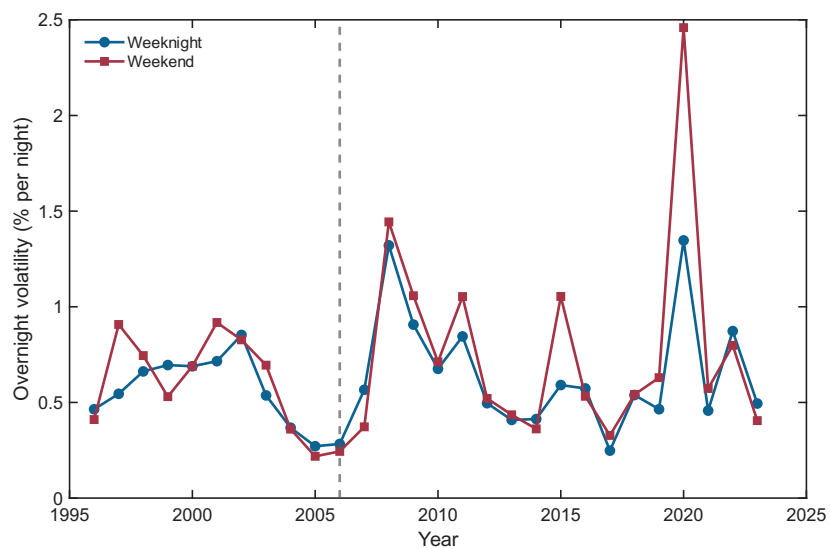
Note: This figure plots the mean difference between intra-week and weekend delta-hedged returns of the out-of-the-money S&P 500 put portfolio of Figure 1 in two-year bins, with 95% Welch (unequal variance) confidence intervals on the difference. Returns are daily close to close, delta-hedged, and in basis points. Weekend observations are Friday-to-Monday returns. The shaded vertical band is centered on January 2006, when overnight equity trading emerged on weeknights but not weekends. The sample period is 1996 to 2023.

Figure IA.3: Alternative Break Dates Yield Similar Difference-in-Differences Estimates



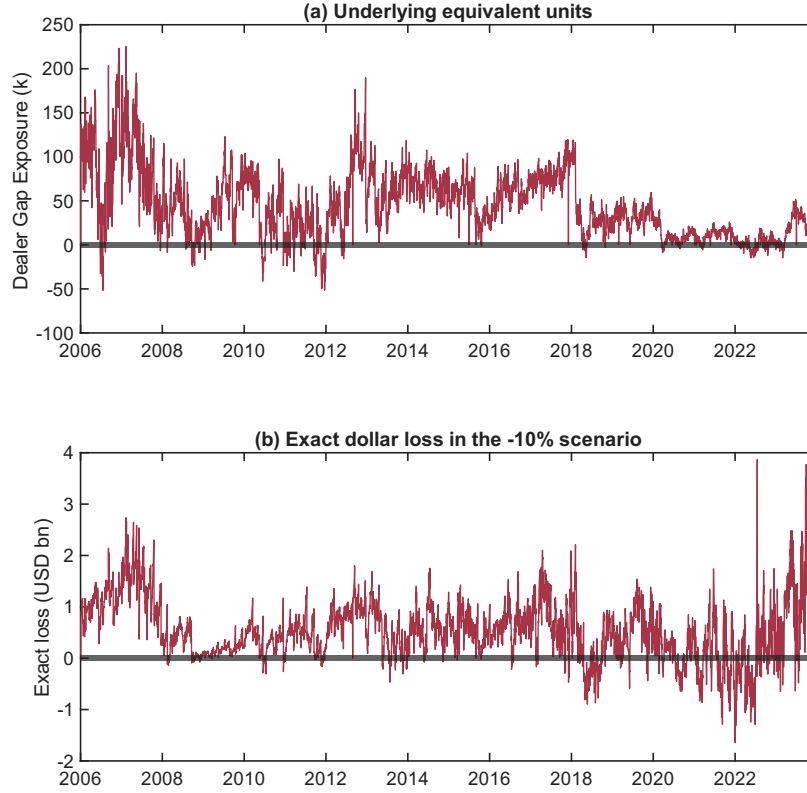
Note: This figure plots the Intra-week $\times$ Post coefficient of Equation (8), reestimated with the break date moved to January of each candidate year from 2002 to 2010. Option returns are daily close-to-close delta-hedged returns of the out-of-the-money S&P 500 put portfolio of Figure 1, in basis points. Whiskers show 95% confidence intervals from Newey–West standard errors with a bandwidth of 22 trading days. The shaded vertical band is centered on January 2006, the break date used throughout the paper. All nine estimates are positive with confidence intervals excluding zero. The sample period is 1996 to 2023.

Figure IA.4: Weeknight and Weekend Volatility Do Not Diverge Around 2006



Note: The figure plots the annual standard deviation of overnight S&P 500 returns, separately for weeknights and weekends. The overnight return is the log change from the prior day's closing level to the next morning's opening level, where the open is measured by the S&P 500 special opening quotation (SOQ). Weekend observations are Friday close to Monday open returns. The vertical line marks January 2006, the emergence of overnight equity trading. Table [IA.5](#) reports the corresponding moments before and after 2006. The sample period is January 1996 to December 2023.

Figure IA.5: Dealer Gap Exposure Declines in Units but Not in Dollars



Note: This figure plots the daily time series of dealers' exposure to gap risk in two panels. Panel (a) plots Dealer Gap Exposure, the negative of the aggregate S&P 500 put book's shadow gamma under a -10% gap (Section I), measured from dealers' net positions (Subsection I.A). Net positions are in underlying equivalent units (contracts times the contract multiplier of 100), and the exposure is reported in thousands. Panel (b) plots the exact loss of the same put book in the -10% scenario, in billions of U.S. dollars. The loss is computed by full revaluation of every position net of the delta hedge, as for the exact exposure of Table IA.15. In both panels, positive values indicate expected inventory losses in the gap. The sample period is January 2006 to December 2023.

IA.5. Additional Tables

Table IA.4: S&P 500 Equity Returns

	Mean	Std	Min	P1	P50	P99	Max
Day	1.5	98.3	-926.2	-271.4	4.7	275.8	817.8
Night (1 day)	2.2	66.5	-749.4	-201.0	4.2	174.7	585.8
Midweek Holiday (2 days)	0.1	80.8	-283.2	-267.1	9.3	307.8	326.6
Weekend (3 days)	-0.4	83.2	-1,235.7	-225.7	3.2	224.2	494.0
Closure of 4+ Days	-3.4	91.8	-462.4	-396.4	3.4	206.3	391.3

Note: This table displays summary statistics for S&P 500 equity returns, in basis points. Returns are measured from S&P 500 index levels. The 09:30 open is the S&P 500 special opening quotation (SOQ), the index level based on the opening auction. Day returns are from 09:30 to 16:00. All other rows are overnight returns from 16:00 to the next trading day's 09:30, split by the calendar length of the market closure. Regular nights span one calendar day, midweek holidays two, and weekends three (Friday 16:00 to Monday 09:30). The last row collects closures of four or more calendar days (holiday weekends and the seven day September 2001 closure). The rows contain 7,018, 5,500, 64, 1,269, and 184 observations, respectively. P1 indicates the first percentile. The sample period is 1996 to 2023.

Table IA.5: Overnight Risk Did Not Change for Weeknights Relative to Weekends Across 2006

	Std (%)	Skew	Freq $ r > 2\%$ (%)	P1 (%)
Weeknight, pre-2006	0.61	-0.32	1.04	-1.76
Weeknight, post-2006	0.71	-0.72	2.15	-2.23
Weekend, pre-2006	0.67	-2.00	2.33	-2.18
Weekend, post-2006	0.92	-3.77	2.95	-2.67

Note: This table reports moments of realized overnight S&P 500 returns for weeknights and weekends, before and after the emergence of overnight equity trading in January 2006. The overnight return is the log change from the prior day's closing level to the next morning's opening level, where the open is measured by the S&P 500 special opening quotation (SOQ), the index level based on the opening auction. Weekend observations are Friday close to Monday open returns. Std is the standard deviation, Freq $|r| > 2\%$ is the share of overnight returns exceeding 2% in absolute value (in percent), and P1 is the first percentile. The sample period is January 1996 to December 2023.

Table IA.6: The Return Rise After 2006 Concentrates in Deep Out-of-the-Money Puts: Full Triple-Differences

	(1) Baseline	(2) Winsorized	(3) Excl. Crashes
Intra-week	85.7** (2.40)	115.1*** (4.41)	85.7** (2.40)
Post	-84.4** (-1.98)	-98.6*** (-3.27)	-154.7*** (-3.89)
Intra-week $\times$ Post	121.9*** (2.60)	92.9*** (2.74)	174.2*** (3.87)
Deep OTM Puts	-307.3*** (-2.94)	-350.7*** (-5.27)	-307.3*** (-2.94)
Intra-week $\times$ Deep OTM Puts	61.7 (0.56)	55.9 (0.73)	61.7 (0.56)
Post $\times$ Deep OTM Puts	-305.6** (-1.97)	-348.1*** (-4.13)	-409.7*** (-3.24)
Intra-week $\times$ Post $\times$ Deep OTM Puts	470.9*** (2.93)	421.3*** (4.38)	561.0*** (3.97)
Constant	-134.7*** (-4.21)	-184.5*** (-7.92)	-134.7*** (-4.21)
Observations	54,842	54,842	53,675
Adjusted R^2	0.00	0.01	0.01

Note: This table reports regression estimates of the triple-difference specification in Equation (9). I regress option returns on indicators for treated options (Deep OTM Puts), treated periods (Intra-week), and the post-event period (Post, observations after January 2006). Option returns are computed for eight out-of-the-money S&P 500 option portfolios (four puts and four calls), formed on lagged absolute delta ($|\Delta| < 0.25$ and $0.25 \leq |\Delta| < 0.5$) and days to expiry (2 to 69 and 70 or more calendar days). Deep OTM Puts indicates the portfolio of deep out-of-the-money puts ($|\Delta| < 0.25$) with 2 to 69 days to expiry. Intra-week returns include all close-to-close observations except Friday-to-Monday intervals. Column (1) presents the baseline regression. Column (2) winsorizes returns at the 5th and 95th percentiles. Column (3) excludes the crisis months from August 2008 to October 2008, February 2018, and from February 2020 to April 2020. Returns are daily close to close, delta-hedged, and in basis points. t -statistics are in parentheses, and standard errors are clustered by month $\times$ portfolio ID. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is 1996 to 2023.

Table IA.7: The 2006 Estimates Are Robust to Inference and Trend Choices

<i>Panel A: The 2006 triple-difference under alternative inference</i>				
	(1) Month $\times$ Port.	(2) Date	(3) Month	(4) Year
Intra-week $\times$ Post $\times$ OTM Puts	470.9*** (2.93)	470.9*** (3.77)	470.9*** (3.79)	470.9*** (3.47)
Intra-week $\times$ Post	121.9*** (2.60)	121.9 (1.57)	121.9 (1.54)	121.9 (1.62)
Intra-week $\times$ OTM Puts	61.7 (0.56)	61.7 (0.67)	61.7 (0.68)	61.7 (0.69)
Post $\times$ OTM Puts	-305.6** (-1.97)	-305.6*** (-2.68)	-305.6*** (-2.65)	-305.6** (-2.29)
Wild bootstrap p (triple)			0.000	0.004
Lower Order Terms	Yes	Yes	Yes	Yes
Clusters	2,656	6,958	332	28
Observations	54,842	54,842	54,842	54,842
Adjusted R^2	0.00	0.00	0.00	0.00

<i>Panel B: Estimates with trends and before the GFC</i>				
	(1) DiD + Trend	(2) DiD to 2007	(3) DDD + Trends	(4) DDD to 2007
Intra-week $\times$ Post $\times$ OTM Puts			768.9*** (2.69)	703.2** (2.23)
Intra-week $\times$ Post	627.5*** (3.95)	570.1*** (3.18)	252.6*** (2.71)	285.7*** (2.99)
Intra-week $\times$ Trend (bps/yr)	-19.7* (-1.93)		-9.5* (-1.68)	
Intra-week $\times$ OTM Puts $\times$ Trend (bps/yr)			-21.5 (-1.11)	
Lower Order Terms	Yes	Yes	Yes	Yes
Observations	6,958	3,015	54,842	23,304
Adjusted R^2	0.01	0.01	0.00	0.00

Note: This table probes the inference and the trend assumptions behind the 2006 difference-in-differences and triple-difference estimates (Table II and Table III). Panel A reestimates the baseline triple-difference of Table III, column (1), whose point estimates are identical by construction, under alternative standard error clustering. Column headers name the cluster dimension, and the Clusters row reports the number of clusters. Two way portfolio and time clustering is degenerate with eight portfolio clusters against eight parameters and is not reported. The Wild bootstrap p row reports wild cluster bootstrap p -values for the triple interaction (Rademacher weights, 999 replications), clustered by month in column (3) and by year in column (4). Panel B reports trend adjusted and pre crisis specifications. Columns (1) and (2) use the difference-in-differences design and sample of Table II (the out-of-the-money put portfolio, with standard errors clustered by month). Columns (3) and (4) use the triple-difference design and sample of Table III (standard errors clustered by month $\times$ portfolio identifier). Trend is calendar time in years, centered at January 2006 and interacted with the treatment indicators, so trend rows read as basis points per year and the Post terms measure the shift at the break date. Columns (2) and (4) end the sample in December 2007, so the post period predates the Global Financial Crisis. Lower order terms are included in every specification. Returns are in basis points and are delta-hedged. t -statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is 1996 to 2023 unless a column states otherwise.

Table IA.8: The Gap Risk Premium Is Robust to Return Measurement Choices

	(1) Baseline	(2) MN Tight	(3) Unhedged	(4) Ask to Bid	(5) Bid to Ask	(6) Trimmed
Night	-174.8*** (-6.12)	-64.9** (-2.00)	-119.2*** (-4.62)	-195.7*** (-7.55)	-239.5*** (-6.34)	-161.3*** (-6.13)
Shadow Gamma	28.6** (2.34)	29.0* (1.78)	-11.2 (-0.58)	-164.6*** (-13.75)	338.6*** (19.32)	25.7** (2.28)
Shadow Gamma $\times$ Night	-158.1*** (-8.66)	-56.5** (-2.29)	-146.7*** (-6.00)	-217.1*** (-12.48)	-61.2*** (-2.64)	-165.9*** (-9.79)
Observations	3,941,285	899,740	3,941,285	3,941,285	3,941,285	3,862,461
Adjusted R^2	0.00	0.00	0.00	0.01	0.01	0.00

Note: This table reestimates the stacked day versus night shadow gamma specification, using day and night delta-hedged option returns on standardized option-level shadow gamma, its interaction with the Night indicator, and no further controls, under alternative samples and return measurement choices. Column (1) is the baseline. It uses mid-quote, delta-hedged returns on the common sample where all return variants are available. The specifications also omit the date and moneyness by maturity bin fixed effects of Table V. The sample restriction and the omitted fixed effects explain the difference from that table's column (1). Column (2) restricts the sample to options with formation $|\Delta| \in (0.10, 0.25)$ and formation mid quotes above \$0.50, the screen of [Muravyev and Ni \(2020\)](#). Column (3) uses unhedged (raw mid-quote) returns. Columns (4) and (5) mark the position legs at executable quotes. Column (4) enters at the ask and exits at the bid, column (5) the reverse. The two conventions shift return levels by the half spread in opposite directions. Column (6) drops (rather than winsorizes) returns beyond the 1st and 99th percentiles. In columns (1) through (5), returns are winsorized at the 2.5th and 97.5th percentiles of the column's estimation sample. Shadow gamma is standardized once on the common sample, so all coefficients are basis points per standard deviation of the same cross section. t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is January 2006 to December 2023.

Table IA.9: The Gap Risk Premium Is Present in Every Moneyness Bin at Executable Prices

Panel A: Delta-hedged returns at mid quotes

	All	Formation $ \Delta $ bin				
		$ \Delta \leq 0.05$	$(0.05, 0.10]$	$(0.10, 0.25)$	$[0.25, 0.50)$	$ \Delta \geq 0.50$
Night	-96.7*** (-3.69)	-451.5*** (-8.10)	-175.3*** (-3.86)	-67.3** (-2.09)	-22.9 (-1.24)	25.4* (1.96)
Shadow Gamma	20.1* (1.75)	12.9 (0.56)	30.3 (1.64)	29.0* (1.78)	42.7*** (2.67)	57.2*** (8.95)
Shadow Gamma $\times$ Night	-106.1*** (-5.95)	-138.0*** (-4.28)	-56.7** (-2.09)	-56.5** (-2.29)	-60.8*** (-2.70)	-50.8*** (-4.91)
Mean shadow gamma (sd units)	-0.02	0.33	0.28	0.04	-0.26	-0.29
SD shadow gamma (sd units)	0.93	1.19	1.17	0.93	0.59	0.70
Observations	3,648,972	591,592	470,416	899,740	1,011,442	675,782
Adjusted R^2	0.00	0.01	0.00	0.00	0.00	0.00

Panel B: Ask to bid marks executable for customers

	All	Formation $ \Delta $ bin				
		$ \Delta \leq 0.05$	$(0.05, 0.10]$	$(0.10, 0.25)$	$[0.25, 0.50)$	$ \Delta \geq 0.50$
Night	-124.6*** (-5.08)	-503.7*** (-10.09)	-303.2*** (-7.21)	-119.2*** (-3.90)	-52.6*** (-2.95)	-4.0 (-0.32)
Shadow Gamma	-150.2*** (-13.14)	-63.0*** (-2.80)	-11.6 (-0.63)	-36.6** (-2.27)	-55.0*** (-3.46)	69.3*** (10.12)
Shadow Gamma $\times$ Night	-128.8*** (-7.46)	-191.4*** (-6.38)	-92.8*** (-3.62)	-66.3*** (-2.79)	-65.1*** (-2.95)	-76.9*** (-7.30)
Observations	3,648,972	591,592	470,416	899,740	1,011,442	675,782
Adjusted R^2	0.01	0.01	0.00	0.00	0.00	0.00

Note: This table decomposes the sample screen of [Muravyev and Ni \(2020\)](#), consisting of formation $|\Delta| \in (0.10, 0.25)$ and formation mid quotes above \$0.50, into its two clauses. Every column restricts the sample to formation mid quotes above \$0.50. The columns then split this executable price sample by formation $|\Delta|$. Column (1) applies no moneyness restriction, and columns (2) through (6) partition it into moneyness bins, with the $(0.10, 0.25)$ bin equal to the full screen. Each column reestimates the stacked day versus night specification of Table [IA.8](#), using day and night delta-hedged option returns on standardized option-level shadow gamma, its interaction with the Night indicator, and no further controls. Panel A marks returns at mid quotes. Panel B enters at the ask and exits at the bid, as in column (4) of Table [IA.8](#). Shadow gamma is standardized once on the common sample of Table [IA.8](#) and centered within each column, so the interaction is basis points per standard deviation of the same cross section in every column, and the Night coefficient equals the column's average night minus day return differential. "Mean shadow gamma" and "SD shadow gamma" report each column's exposure distribution in the same standardized units. Returns are winsorized at the 2.5th and 97.5th percentiles of the column's estimation sample. t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is January 2006 to December 2023.

Table IA.10: The Gap Risk Premium Strengthens Under Minimum-Variance Hedging

	(1) BSM	(2) MV	(3) Unhedged	(4) BSM, Win.	(5) MV, Win.	(6) Unh., Win.
Night	-281.2*** (-5.55)	-294.6*** (-6.10)	-250.8*** (-5.00)	-198.7*** (-6.56)	-198.7*** (-6.36)	-143.1*** (-5.19)
Shadow Gamma	32.7* (1.66)	52.9** (2.52)	-11.1 (-0.36)	30.0** (2.35)	56.8*** (3.76)	-10.4 (-0.52)
Shadow Gamma $\times$ Night	-176.7*** (-6.62)	-198.9*** (-6.84)	-169.0*** (-4.69)	-168.4*** (-8.85)	-180.1*** (-8.21)	-156.2*** (-6.16)
Night mean (bps)	-252.9	-220.9	-226.4	-283.3	-228.7	-329.9
Observations	3,693,984	3,693,984	3,693,984	3,693,984	3,693,984	3,693,984
Adjusted R^2	0.00	0.00	0.00	0.00	0.00	0.00

Note: This table reestimates the stacked day versus night shadow gamma specification, using day and night option returns on standardized option-level shadow gamma and its interaction with the Night indicator, under three hedge conventions on one common sample. The conventions are the baseline Black–Scholes–Merton delta-hedge (columns 1 and 4), a minimum-variance delta-hedge (columns 2 and 5), and no hedge (columns 3 and 6). The minimum-variance delta adds to the Black–Scholes–Merton delta the vega times the empirical smile slope, $\partial\sigma_{IV}/\partial K$, estimated per date, session leg, and expiry, separately for out-of-the-money puts and calls, from cells with at least four contracts (slopes winsorized at the 1st and 99th percentiles). Because the index smile slopes downward, the minimum-variance delta lies below the Black–Scholes–Merton delta. The “Night mean” row reports the average night return per convention. Columns (4) through (6) winsorize returns at the 2.5th and 97.5th percentiles of the common sample. The common sample loses about 6% of the stacked day versus night estimation sample (rows without a four contract smile fit). t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample period is January 2006 to December 2023.

Table IA.11: Dealer Gap Exposure Is Priced in the Put Book

	Winsorized			Unwinsorized		
	(1) Puts	(2) Calls	(3) All options	(4) Puts	(5) Calls	(6) All options
Night	-186.5*** (-4.13)	-61.7 (-0.80)	-139.6*** (-4.05)	-286.2*** (-4.38)	-158.0 (-1.56)	-238.9*** (-4.20)
Dealer Gap Exposure $\times$ Night	-145.4*** (-3.74)	-39.0 (-0.90)	-108.7*** (-3.88)	-236.2*** (-2.74)	-73.8 (-1.27)	-189.7** (-2.30)
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,045,060	1,221,981	3,267,041	2,045,060	1,221,981	3,267,041
Adjusted R^2	0.29	0.28	0.19	0.21	0.24	0.17

Note: This table estimates the baseline specification of Table IV, column (1), segment by segment, matching each side of the dealer inventory to its own market segment. Column (1) regresses day and night delta-hedged returns of out-of-the-money S&P 500 puts on the put book's Dealer Gap Exposure and reproduces Table IV, column (1). Column (2) regresses out-of-the-money call returns on the call book's exposure. Column (3) regresses puts and calls jointly on the exposure of the full book. Columns (4) through (6) repeat the three specifications with unwinsorized returns. Each column's Dealer Gap Exposure is measured from dealers' net positions (Subsection I.A) at the start of the night and standardized to zero mean and unit variance within that column's estimation sample. Higher values indicate a more exposed book. Option returns are measured in basis points. In columns (1) through (3) they are winsorized at the 2.5th and 97.5th percentiles of each column's estimation sample. All columns include date fixed effects and moneyness by maturity bin fixed effects built within segment (twenty moneyness quantiles by five maturity buckets, interacted with option type in columns (3) and (6)). t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. In column (4) the double clustered variance matrix is not positive semi-definite, and standard errors are clustered by trading day. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample runs from January 2006 through December 2023 and contains 2,045,060 put, 1,221,981 call, and 3,267,041 pooled day and night observations.

Table IA.12: The Night Price of Gap Exposure Holds Under Date-Level Aggregation, Exact Matching, and a Predetermined Book

	Date level		Matched pairs		Predetermined book	
	(1)	(2)	(3)	(4)	(5)	(6)
Dealer Gap Exposure ($t - 1$)	-93.9*** (-4.18)	-104.7*** (-3.17)	-187.4*** (-4.07)	-126.7*** (-4.78)		
Dealer Gap Exposure ($t - 2$) $\times$ Night					-130.2*** (-3.31)	-150.2*** (-2.59)
Night					-186.3*** (-4.12)	-187.3*** (-4.13)
Constant	-205.0*** (-6.22)	-201.9*** (-4.78)	-119.5** (-2.19)	-161.1*** (-4.02)	-115.2*** (-8.56)	-115.6*** (-8.67)
Unit	Date	Date	Pair	Date	Option	Option
Inference	NW(22)	NW(22)	Date cl.	NW(22)	Date cl.	2-way cl.
Vol. controls and trend	No	Yes	No	No	No	Yes
Moneyiness $\times$ Maturity FE	—	—	—	—	Yes	Yes
Date FE	—	—	—	—	Yes	Yes
Dates	4,472	4,472	4,471	4,471	4,509	4,509
Observations	4,472	4,472	337,737	4,471	2,045,060	2,045,060
Adjusted R^2	0.003	0.004	0.004	0.004	0.294	0.294

Note: This table reestimates the price of dealer gap exposure of Table IV under three designs. Columns (1) and (2) collapse the sample to one observation per trading day, the difference between the equal weighted mean night return and the mean day return, and regress it on Dealer Gap Exposure. Column (2) adds the lagged VIX, the day versus night volatility ratio, the variance risk premium, and a linear time trend. Standard errors are Newey–West with 22 lags. Columns (3) and (4) use only exactly matched observations, the same option on the same date with a valid return in both sessions. Column (3) regresses the within option night minus day return difference on Dealer Gap Exposure with standard errors clustered by trading day, and column (4) collapses the matched differences to date means with Newey–West standard errors (22 lags). Columns (5) and (6) reestimate Table IV, columns (1) and (4), with the book measured one further trading day back, where $t - 1$ denotes the close at which the night begins, the start of night state used throughout the paper, and $t - 2$ the close one trading day earlier. The correlation between the two book states is 0.922. Standard errors in column (6) are double clustered by trading day and option identifier, and in column (5) are clustered by trading day, the double clustered variance matrix not being positive semi-definite there. Dealer Gap Exposure is measured from dealers’ net positions (Subsection I.A) and standardized to zero mean and unit variance within the estimation sample of Table IV, so slopes are comparable across columns and to that table. Option returns are in basis points and winsorized at the 2.5th and 97.5th percentiles of that sample. In columns (1), (2), and (4) the adjusted R^2 is from the corresponding least squares fit. t -statistics are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample runs from January 2006 through December 2023 and contains 4,472 dates with both sessions, 337,737 matched pairs (4,471 of the 4,472 dates contain at least one matched pair), and 2,045,060 stacked day and night observations.

Table IA.13: The Night Price of Gap Exposure Holds on the Directly Reported Market Maker Positions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Night	-180.4*** (-3.74)	-180.7*** (-3.75)	-180.3*** (-3.75)	-180.7*** (-3.76)	-164.2*** (-3.20)	-178.6*** (-3.73)	-168.7*** (-3.49)
Dealer Gap Exposure $\times$ Night	-148.6*** (-3.40)	-141.3** (-2.52)	-141.1** (-2.49)	-199.6*** (-2.99)	-145.1** (-2.19)	-174.8** (-2.47)	-191.7*** (-2.87)
Vol. controls ($\times$ Night)	No	VIX	Full	Full	Full+	Full	Full+
Trend ($\times$ Night)	No	No	No	Yes	Yes	Yes	Yes
Option chars. ($\times$ Night)	No	No	No	No	Yes	No	No
Dealer Gamma, Vanna ($\times$ Night)	No	No	No	No	No	Yes	No
Interm. constraints ($\times$ Night)	No	No	No	No	No	No	Yes
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,924,634	1,924,634	1,924,634	1,924,634	1,649,291	1,924,634	1,924,634
Adjusted R^2	0.30	0.30	0.30	0.30	0.31	0.30	0.30

Note: This table repeats Table IV, with all seven columns specified identically, on the direct market maker position reports. Beginning in January 2011, the CBOE Open-Close files flag market maker trades explicitly, and Dealer Gap Exposure is built from those flagged trades alone rather than from the residual dealer net flows of Subsection I.A. At the date level, the two put book exposure measures correlate 0.931 over the overlap. The window starts after a half-year burn-in of the cumulated book. All variables are standardized within this table's estimation sample. t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample contains 1,924,634 day and night observations from July 2011 to December 2023. Column (5) requires nonmissing characteristics and contains 1,649,291 observations.

Table IA.14: The Night Price of Gap Exposure Strengthens Within Years and Within Months

	(1)	(2)	(3)	(4)	(5)	(6)
Night	-186.5*** (-4.13)			-187.4*** (-4.13)		
Dealer Gap Exposure $\times$ Night	-145.4*** (-3.74)	-231.5*** (-3.88)	-223.9** (-2.41)	-187.0*** (-3.37)	-246.2*** (-3.78)	-209.5** (-2.29)
Calendar $\times$ Night FE	None	Year	Month	None	Year	Month
Vol. controls ($\times$ Night)	No	No	No	Full	Full	Full
Trend ($\times$ Night)	No	No	No	Yes	No	No
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,045,060	2,045,060	2,045,060	2,045,060	2,045,060	2,045,060
Adjusted R^2	0.29	0.30	0.31	0.29	0.30	0.31

Note: This table estimates the specification of Table IV with calendar $\times$ Night fixed effects, which give each calendar period its own night return level, so that the exposure slope is identified from variation of the book within the period. Columns (1) and (4) reproduce columns (1) and (4) of Table IV. These are the baseline and the specification with the volatility controls and a linear time trend interacted with Night. Columns (2) and (5) add year $\times$ Night fixed effects. Columns (3) and (6) add month $\times$ Night fixed effects. In these four columns the Night level is absorbed, and in columns (5) and (6) the fixed effects replace the linear trend term. All regressors are standardized within the estimation sample. t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. The sample contains 2,045,060 day and night observations of out-of-the-money S&P 500 puts from January 2006 to December 2023, identical in every column.

Table IA.15: The Price of Gap Risk Holds with the Exact Measure of Shadow Gamma

	Dealer book				Option level
	(1)	(2)	(3)	(4)	(5)
Night	-182.1*** (-3.98)	-184.1*** (-4.07)	-182.2*** (-3.99)	-182.6*** (-4.00)	-104.0*** (-3.50)
Exact Dealer Gap Exposure $\times$ Night	-118.8*** (-4.24)	-156.9*** (-3.96)	-93.4*** (-3.69)	-135.1*** (-4.34)	
Exact Shadow Gamma per Dollar of Option Value $\times$ Night					-186.5*** (-7.12)
Gap scenario	-10%	-10%	-5%	-15%	-10%
Vol. controls ($\times$ Night)	No	Full	No	No	No
Trend ($\times$ Night)	No	Yes	No	No	No
Moneyness $\times$ Maturity FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Observations	2,045,060	2,045,060	2,045,060	2,045,060	3,935,642
Adjusted R^2	0.29	0.29	0.29	0.29	0.17

Note: This table repeats the paper’s principal regressions with the exact shadow gamma of Equation (4) in place of the proxy of Equation (5). Columns (1) through (4) estimate the specification of Table IV. Dealer Gap Exposure is recomputed from each position’s exact shadow gamma, so that Equation (6) aggregates the full path of gamma rather than gamma at the shocked underlying level. The Gap scenario row gives the hypothetical underlying return of each column. Because the exact exposure varies only by date, its level is absorbed by the date fixed effects. Column (2) adds the lagged VIX, λ , the lagged day versus night volatility ratio, the lagged variance risk premium, and a linear time trend, each interacted with the Night indicator, as in column (4) of Table IV. Column (5) estimates the specification of column (1) of Table V, replacing shadow gamma with the exact measure per dollar of option value, the option’s exact delta-hedged revaluation under a -10% gap divided by its price. Like the proxy it replaces, the revaluation is measured at the previous close for both sessions and enters with its level. It is winsorized at the 2.5th and 97.5th percentiles of the estimation sample before standardization. Returns are in basis points, winsorized at the 2.5th and 97.5th percentiles of each estimation sample. All regressors are standardized within the estimation sample, so slopes are basis points per standard deviation. All columns include date fixed effects and moneyness by maturity bin fixed effects (twenty moneyness quantiles by five maturity buckets, formed within option type in column (5)). t -statistics are in parentheses, and standard errors are double clustered by trading day and option identifier. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Columns (1) through (4) contain 2,045,060 day and night observations of out-of-the-money S&P 500 puts from January 2006 to December 2023. Column (5) contains 3,935,642 day and night observations of all S&P 500 options over the same window (fourteen observations of the Table V sample lack the exact revaluation).